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Systems, methods, and apparatus for dividing and compressing data

Abstract

A method for data compression may include scanning input data, performing, based on the scanning, a compression operation to generate compressed data using the input data, finding, based on the scanning, a delimiter in the input data, and generating, based on a position of the delimiter in the input data, a portion of data using the compressed data. The input data may include a record, the delimiter indicates a boundary of the record, and the portion of data may include the record. The generating may include generating the portion of data based on a portion size. The portion size may be a default portion size. The portion size may be based on a default portion size and a length of a match in the input data.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to, and the benefit of, U.S. Provisional Patent Application Ser. No. 63/231,709 titled “Object Processing and Filtering for Computational Storage” filed Aug. 10, 2021 which is incorporated by reference, U.S. Provisional Patent Application Ser. No. 63/231,711 titled “Data Placement with Spatial Locality and Hierarchical Aggregation for Computational Storage” filed Aug. 10, 2021, which is incorporated by reference, U.S. Provisional Patent Application Ser. No. 63/231,710 titled “Integrated Data Chunking and Compression for Near Data Processing” filed Aug. 10, 2021 which is incorporated by reference, and U.S. Provisional Patent Application Ser. No. 63/231,715 titled “Integrated Data Chunking and Encryption for Near Data Processing” filed Aug. 10, 2021 which is incorporated by reference.

TECHNICAL FIELD

(1) This disclosure relates generally to computational storage systems, and more specifically to systems, methods, and apparatus for dividing and compressing data.

BACKGROUND

(2) A computational storage device may include one or more processing resources that may operate on data stored at the device. A host may offload a processing task to the storage device, for example, by sending a command to the storage device indicating an operation to perform on data stored at the device. The storage device may use the one or more processing resources to execute the command. The storage device may send a result of the operation to the host and/or store the result at the device.

(3) The above information disclosed in this Background section is only for enhancement of understanding of the background of the inventive principles and therefore it may contain information that does not constitute prior art.

SUMMARY

(4) A method for data compression may include receiving input data, finding a delimiter in the input data, generating, based on a position of the delimiter in the input data, a portion of data using

a part of the input data, and compressing the portion of data. The input data may include a record, the delimiter indicates a boundary of the record, and the portion of data may include the record. The position of the delimiter may be in the part of the input data. Generating the portion of data may include generating the portion of data based on a subset of the part of the input data. The part of the input data may be a first part of the input data, and the position of the delimiter may be in a second part of the input data. Generating the portion of data may include generating the portion of data based on the first part of the input data and the second part of the input data. The size of the part of input data may be based on a default portion size. The method may further include modifying a size of the portion of data based on the position of the delimiter. Modifying the size of the portion of data may include extending the size of the portion of data. The receiving may include receiving a stream of input data. Finding a delimiter in the input data may include performing a first scan operation on the input data, and compressing the portion of data may include performing a second scan operation on the portion of data.

(5) A method for data compression may include scanning input data, performing, based on the scanning, a compression operation to generate compressed data using the input data, finding, based on the scanning, a delimiter in the input data, and generating, based on a position of the delimiter in the input data, a portion of data using the compressed data. The input data may include a record, the delimiter indicates a boundary of the record, and the portion of data may include the record. The generating may include generating the portion of data based on a portion size. The portion size may be a default portion size. The portion size may be based on a default portion size and a length of a match in the input data. The match may include the delimiter. The delimiter may be a first delimiter, the method may further include keeping the first delimiter, and finding a second delimiter in the input data, wherein the generating may include generating the portion of data based on the first delimiter and the second delimiter. The method may further include setting an indication based on a size of the compressed data. The indication may include a termination indication. The generating may include generating the portion of data based on the indication and the delimiter. The scanning may include scanning the input data based on the indication. The scanning may include scanning the input data based on the delimiter. The performing may include performing the compression operation based on the delimiter. The input data may include a stream of input data. The compression operation may include a stream-based compression operation.

(6) A system may include a host comprising host logic configured to perform a scanning operation on input data, perform, based on the scanning operation, a data compression operation to generate compressed data using the input data, find, based on the scanning operation, a delimiter in the input data, and generate, based on the delimiter, a portion of data using the compressed data. The host logic may be further configured to generate the portion of data based on a default portion size and a position of the delimiter in the input data. The delimiter may be a first delimiter, and the host logic may be further configured to generate the portion of data based on a position of a second delimiter in the input data. The system may further include a device configured to receive the portion of data from the host, the device comprising device logic configured to decompress the portion of data to generate a decompressed portion of data, and perform an operation on the decompressed portion of data.

(7) A method for data encryption may include receiving input data, finding a delimiter in the input data, generating, based on a position of the delimiter in the input data, a portion of data using a part of the input data, and encrypting the portion of data. The input data may include a record, the delimiter indicates a boundary of the record, and the portion of data may include the record. The position of the delimiter may be in the part of the input data. Generating the portion of data may include generating the portion of data based on a subset of the part of the input data. The part of the input data may be a first part of the input data, and the position of the delimiter may be in a second part of the input data. Generating the portion of data may include generating the portion of data based on the first part of the input data and the second part of the input data. The size of the part of

input data may be based on a default portion size. The method may further include modifying a size of the portion of data based on the position of the delimiter. Modifying the size of the portion of data may include extending the size of the portion of data. Receiving input data may include receiving a stream of input data. Finding a delimiter in the input data may include performing a first scan operation on the input data, and encrypting the portion of data may include performing a second scan operation on the portion of data.

(8) A method for data encryption may include scanning input data, performing, based on the scanning, an encryption operation to generate encrypted data using the input data, finding, based on the scanning, a delimiter in the input data, and generating, based on a position of the delimiter in the input data, a portion of data using the encrypted data. The input data may include a record, the delimiter indicates a boundary of the record, and the portion of data may include the record. The generating may include generating the portion of data based on a portion size. The portion size may be a default portion size. The generating may include extending a size of the portion of data based on the position of the delimiter in the input data. The encryption operation may include a block-based encryption operation, and the performing may include performing the encryption operation on a first block of the input data to generate a first block of encrypted data. The delimiter may be located in a second block of input data, the method further comprising extending a size of the portion of data based on the position of the delimiter. The method may further include extending the size of the portion of data based on a size of the second block of input data. The extending may include padding the portion of data based on the size of the second block of input data. The size of the second block of input data may be based on a key length for the encryption operation. The performing may further include performing the encryption operation on the second block of the input data to generate a second block of encrypted data. The generating may include generating the portion of data using the first block of encrypted data and the second block of encrypted data.

(9) A system may include a host comprising host logic configured to perform a scanning operation on input data, perform, based on the scanning operation, a data encryption operation to generate encrypted data using the input data, find, based on the scanning operation, a delimiter in the input data, and generate, based on the delimiter, a portion of data using the encrypted data. The host logic may be further configured to generate the portion of data based on a default portion size and a position of the delimiter in the input data. The host logic may be further configured to generate the portion of data based on a key length of the encryption operation. The system may further include a device configured to receive the portion of data from the host, the device comprising device logic configured to decrypt the portion of data to generate a decrypted portion of data, and perform an operation on the decrypted portion of data.

(10) A method of dividing data may include scanning input data, performing, based on the scanning, an operation using the input data to generate processed data, finding, based on the scanning, a delimiter in the input data, and generating, based on a position of the delimiter in the input data, a portion of data using the processed data. The operation may include a data compression operation. The operation may include a data encryption operation. The generating may include generating the portion of data based on a default portion size. The input data may include a stream of input data, and the operation may include a streaming-based operation. The operation may include a block-based operation. The performing may include performing the operation using a block of the input data. The generating may include modifying the portion of data based on the position of the delimiter in the input data. The modifying may include extending the portion of data. The extending may include extending the portion of data based on a block size of the operation. The extending may include padding the portion of data.

(11) A system may include a host comprising host logic configured to perform a scanning operation on input data, perform, based on the scanning operation, a processing operation to generate processed data using the input data, find, based on the scanning operation, a delimiter in the input data, and generate, based on the delimiter, a portion of data using the processed data. The host logic

may be further configured to generate the portion of data based on a default portion size and a position of the delimiter in the input data. The system may further include a device configured to receive the portion of data from the host, the device comprising device logic configured to restore the portion of data to generate a restored portion of data, and perform an operation on the restored portion of data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The figures are not necessarily drawn to scale and elements of similar structures or functions may generally be represented by like reference numerals or portions thereof for illustrative purposes throughout the figures. The figures are only intended to facilitate the description of the various embodiments described herein. The figures do not describe every aspect of the teachings disclosed herein and do not limit the scope of the claims. To prevent the drawings from becoming obscured, not all of the components, connections, and the like may be shown, and not all of the components may have reference numbers. However, patterns of component configurations may be readily apparent from the drawings. The accompanying drawings, together with the specification, illustrate example embodiments of the present disclosure, and, together with the description, serve to explain the principles of the present disclosure.
- (2) FIG. 1A illustrates an embodiment of an object storage scheme with server-side encryption in accordance with example embodiments of the disclosure.
- (3) FIG. 1B illustrates an embodiment of an object storage scheme with client-side encryption in accordance with example embodiments of the disclosure.
- (4) FIG. 2A illustrates an embodiment of an object storage scheme that may return an object to a user's device in accordance with example embodiments of the disclosure.
- (5) FIG. 2B illustrates an embodiment of an object storage scheme having a data selection feature in accordance with example embodiments of the disclosure.
- (6) FIG. 3A illustrates an embodiment of a write operation of an object storage scheme having a data selection feature in accordance with example embodiments of the disclosure.
- (7) FIG. 3B illustrates an embodiment of a read operation of an object storage scheme having a data selection feature in accordance with example embodiments of the disclosure.
- (8) FIG. 4 illustrates an embodiment of a storage system having local data restoration in accordance with example embodiments of the disclosure.
- (9) FIG. 5 illustrates another embodiment of a storage system having local data restoration in accordance with example embodiments of the disclosure.
- (10) FIG. 6A illustrates an example embodiment of a write operation for a storage scheme having local data restoration and server-side encryption in accordance with example embodiments of the disclosure.
- (11) FIG. 6B illustrates an example embodiment of a write operation for a storage scheme having local data restoration and client-side encryption in accordance with example embodiments of the disclosure.
- (12) FIG. 7A illustrates an example embodiment of a write operation for a storage scheme having local data restoration in accordance with example embodiments of the disclosure.
- (13) FIG. 7B illustrates an example embodiment of a read operation with data selection for a storage scheme having local data restoration in accordance with example embodiments of the disclosure.
- (14) FIG. 8 illustrates an example embodiment of a system architecture for an object storage scheme with local data restoration in accordance with example embodiments of the disclosure.
- (15) FIG. 9A illustrates example embodiments of read and/or write operations for a storage scheme

with local data restoration in accordance with example embodiments of the disclosure.

(16) FIG. 9B illustrates an example embodiment of a read operation for a storage scheme with local data restoration and a data selection operation in accordance with example embodiments of the disclosure.

(17) FIG. 10 illustrates an embodiment of a distribution of the data from Table 1 across three data chunks at three computational storage devices in accordance with example embodiments of the disclosure.

(18) FIG. 11 illustrates an example embodiment of a storage system in which a server may reconstruct records split between data chunks at different storage devices in accordance with example embodiments of the disclosure.

(19) FIG. 12 illustrates an embodiment of a content-aware method for dividing data into chunks and compressing data in accordance with example embodiments of the disclosure.

(20) FIG. 13 illustrates an embodiment of a method for compressing data in accordance with example embodiments of the disclosure.

(21) FIG. 14 illustrates an embodiment of operations that may be performed by the method illustrated in FIG. 13.

(22) FIG. 15 illustrates an embodiment of an integrated chunking and compression scheme in accordance with example embodiments of the disclosure.

(23) FIG. 16 illustrates an embodiment of an integrated chunking and compression method that may be performed by the scheme illustrated in FIG. 15 in accordance with example embodiments of the disclosure.

(24) FIG. 17 illustrates another embodiment of an integrated chunking and compression scheme in accordance with example embodiments of the disclosure.

(25) FIG. 18 illustrates an embodiment of an integrated chunking and compression method that may be performed by the scheme illustrated in FIG. 17 in accordance with example embodiments of the disclosure.

(26) FIG. 19 illustrates another embodiment of an integrated chunking and compression method in accordance with example embodiments of the disclosure.

(27) FIG. 20 illustrates an embodiment of a system with integrated chunking and compression in accordance with example embodiments of the disclosure.

(28) FIG. 21 illustrates an embodiment of a content-aware method for dividing data into chunks and encrypting data in accordance with example embodiments of the disclosure.

(29) FIG. 22 illustrates an example embodiment of a block cipher method that may be used to encrypt data techniques in accordance with example embodiments of the disclosure.

(30) FIG. 23 illustrates an embodiment of an integrated data chunking and encryption scheme in accordance with example embodiments of the disclosure.

(31) FIG. 24 illustrates an embodiment of an integrated data chunking and encryption method in accordance with example embodiments of the disclosure.

(32) FIG. 25 illustrates an embodiment of a system with integrated chunking and encryption in accordance with example embodiments of the disclosure.

(33) FIG. 26 illustrates an example embodiment of a host apparatus for a storage scheme with data chunk modification in accordance with example embodiments of the disclosure.

(34) FIG. 27 illustrates an example embodiment of a storage device with data chunk modification in accordance with example embodiments of the disclosure.

(35) FIG. 28 illustrates an embodiment of a method for data compression in accordance with example embodiments of the disclosure.

(36) FIG. 29 illustrates an embodiment of a method for data encryption in accordance with example embodiments of the disclosure.

(37) FIG. 30 illustrates an embodiment of a method for data chunking data in accordance with example embodiments of the disclosure.

DETAILED DESCRIPTION

(38) An object storage system may implement a data selection feature that may enable a user's device to request a specified subset of data to retrieve from a stored object. To process such a request, a storage server may reconstruct the object from one or more portions of data stored on one or more storage devices. The storage server may also decrypt the object if it was encrypted, and/or decompress the object if it was compressed to restore the object to its original form. The storage server may perform one or more selection operations such as filtering, scanning, and/or the like, on the restored object to find the specified subset of data requested by the user's device. The storage server may return the requested subset of data to the user's device.

(39) In some respects, a computational storage device may be capable of performing one or more selection operations such as filtering, scanning, and/or the like, on an object stored on the device. However, if a portion of the object is stored on the device, and the object was modified (e.g., compressed, encrypted, and/or the like) prior to dividing the data into portions, the portion stored on the device may only include random (to the device) information that the storage device may not be able to restore (e.g., decompress and/or decrypt) to original data. Therefore, the storage device may not be able to perform a meaningful operation locally on the portion of data stored at the device.

(40) This disclosure encompasses numerous principles relating to computational storage. The principles disclosed herein may have independent utility and may be embodied individually, and not every embodiment may utilize every principle. Moreover, the principles may also be embodied in various combinations, some of which may amplify some benefits of the individual principles in a synergistic manner.

(41) Some of the principles disclosed herein relate to dividing data into one or more portions prior to performing one or more modifications on the one or more portions. For example, in a computational storage scheme in accordance with example embodiments of the disclosure, an object or other original data may be divided into portions of data prior to performing modifications such as compression and/or encryption on the data. One or more of the portions of data may be modified individually (e.g., compression and/or encryption may be performed on an individual portion of the data), and the modified version of the portion of data may be sent to a computational storage device for storage and/or processing. The storage device may generate a restored version of the portion of data from the modified portion of data, for example, by decrypting and/or decompressing the modified portion of data. The storage device may perform an operation (e.g., a selection operation) locally on the restored portion of data.

(42) Depending on the implementation details, performing a selection operation locally at a computational storage device may reduce the amount of data that may be sent from one or more storage devices to a server. Moreover, depending on the implementation details, a computational storage device may perform an operation such as a selection operation more efficiently than a server.

(43) In some example embodiments in accordance with the disclosure, a storage device, a storage server, and/or the like, may provide one or more indications of how to divide original data into portions and/or how to modify the portions to facilitate storage and/or processing by one or more computational storage devices. For example, in some embodiments, an indication may include information such as one or more portion sizes, compression algorithms, encryption algorithms, and/or the like, that may be supported by a storage device. In some embodiments, one or more indications may be mandatory, optional (e.g., advisory), or a combination thereof. For example, an indication of an optimal portion size for storage on a particular storage device may be advisory, whereas an indication of a supported compression algorithm may be mandatory to enable a storage device to decompress a portion of data for local processing at the device.

(44) Any of the operations disclosed herein including dividing data, modifying data (e.g., compressing and/or encrypting data), erasure coding data, storing data, processing data, selecting

data, and/or the like, may be distributed (e.g., mapped) among various apparatus in unlimited configurations in accordance with example embodiments of the disclosure. For example, in some embodiments, a client may divide original data (e.g., an object) into one or more portions, compress the portions of data, and send the compressed portions of data to a server. The server may encrypt the compressed portions of data, and store the compressed and encrypted portions of data across one or more storage devices. As another example, in some embodiments, a client may divide original data (e.g., an object) into one or more portions, compress and encrypt the portions of data, and send the compressed and encrypted portions of data to a server for storage across one or more storage devices. As a further example, a client may send original data (e.g., an object), to a server which may divide the data into one or more portions, and compress, encrypt, and/or perform erasure coding on the portions of data, and store the individually modified portions of data across one or more storage devices.

(45) Some additional principles of this disclosure relate to content-aware techniques for dividing data into portions. For example, in some embodiments, a portion size may be determined dynamically by analyzing the contents of a portion to find a boundary (indicated, for example, by a delimiter) of a record within the data being divided. The portion size may be determined to align with one or more complete records within the portion. For example, if a portion of data having a default portion size includes a partial record, the size of the portion may be modified (e.g., extended or reduced) so the portion ends with a delimiter of a record within the portion (e.g., the portion may only include complete records). The resulting self-contained portion may be compressed and/or encrypted as a unit.

(46) Some additional content-aware data dividing techniques in accordance with example embodiments of the disclosure may integrate a data dividing operation with another operation that may scan the data to be divided. For example, in some embodiments, a data dividing operation may be combined with a data compression operation such that, as a stream of input data is being scanned for purposes of compression, it may also be scanned for one or more delimiters that indicate one or more boundaries of one or more records. A data portion size may be determined by the location of one or more delimiters so the portion ends at the end of a record. In some embodiments, the data compression operation may also terminate at the end of the portion. Depending on the implementation details, this may improve the efficiency of the data dividing operation because it may exploit the data scanning that was already being performed for purposes of compression.

(47) As another example, in some embodiments, a data dividing operation may be combined with a data encryption operation. As a stream of input data is being read for purposes of encryption, it may also be scanned for one or more delimiters that indicate one or more boundaries of one or more records. A data portion size may be determined by the location of one or more delimiters so the portion ends at the end of a record. In some embodiments, the data encryption operation may also terminate at the end of the portion. If the encryption operation is implemented with a block-cipher algorithm, the size of a portion may be modified (e.g., extended or reduced) so the end of the portion aligns with the end of a block (which may be padded if the size of the data is not an even multiple of the block size).

(48) In some embodiments, a portion of data may also be referred to as a chunk of data, and dividing data into portions or chunks of data may be referred to as chunking data. In some embodiments, a portion or chunk of data may refer to any unit of data that may be obtained by dividing data, for example, for purposes of storage at one or more storage devices. In some situations, if an amount of original data is less than or equal to a portion or chunk size (e.g., a default portion or chunk size) a unit of the original data generated by a dividing or chunking operation may still be referred to as a portion or chunk of data, even if it is the same size as the amount of original data.

(49) For purposes of illustration, some embodiments may be described in the context of object

storage systems that may implement a data selection feature and/or may store data in one or more key-value (KV) storage devices. However, the principles described in this disclosure are not limited to any particular data format, data processing features, storage device interfaces, and/or the like. For example, systems, methods, and/or apparatus in accordance with example embodiments of the disclosure may also be implemented with storage systems that may provide file storage, database storage, block storage, and/or the like, may implement any type of processing features such as acceleration, graph processing, graphics processing, machine learning, and/or the like, and may operate with any type of storage devices including KV storage devices, block storage devices, and/or the like.

(50) An object storage system may enable a user's device to store data in the form of objects. The data in an object may be modified in various ways prior to being stored. For example, the data may be compressed to reduce the amount of space it occupies in storage media and/or to reduce the time, bandwidth, power, and/or the like, required to transmit the data from a client to one or more storage devices (e.g., over a network). As another example, the data in an object may be encrypted to prevent unauthorized access to the data during transmission and/or storage of the data.

(51) An object may include a relatively large amount of data, and thus, for purposes of reliability, accessibility, and/or the like, the object may be divided into chunks that may be stored across multiple storage devices. (Dividing data into chunks may also be referred to as chunking the data.) For example, after compression and/or encryption, an object may be divided into fixed-size chunks to fit in a block size used by one or more block-based storage devices in the storage system. In some embodiments, an erasure coding scheme may be used to divide the data into data chunks and generate one or more parity chunks that may enable a storage system to recover a lost or corrupted data chunk.

(52) FIG. 1A illustrates an embodiment of an object storage scheme with server-side encryption in accordance with example embodiments of the disclosure. The left side of FIG. 1A illustrates data flow between components of a system during read and/or write operations, and the right side of FIG. 1A illustrates operations on data during a write operation.

(53) The system illustrated on the left side of FIG. 1A may include a client **102**, one or more servers **104** (which may be referred to collectively as a server), and one or more storage devices **108** (which may be referred to collectively as storage). The operations illustrated on the right side of FIG. 1A are shown in a first group **110A** performed by the client **102** and a second group **112A** performed by the server **104**.

(54) During a write operation, the client **102** may begin with original data **114** which may be, for example, an object. The client **102** may perform one or more compression operations on the original data **114** to generate compressed data **116**. The client **102** may send the compressed data **116** to the server **104** which may encrypt the compressed data **116** to generate encrypted data **118**. The server **104** may divide the compressed and encrypted data **118** into one or more data chunks **120** and send the one or more data chunks **120** to one or more storage devices **108**. In some embodiments, the server **104** may perform erasure coding on the one or more data chunks **120** to generate one or more parity chunks **121** which may also be stored on the one or more storage devices **108**.

(55) During a read operation, the operations shown in FIG. 1A may be performed in reverse. For example, the server **104** may read the one or more data chunks **120** from the one or more storage devices **108**. If one of the data chunks is missing or corrupted, for example, due to a failed storage device, the server **104** may recover the missing or corrupted data chunk using the one or more parity chunks **121**. The server **104** may reconstruct the compressed and encrypted data **118** from the data chunks **120**. The server **104** may decrypt the compressed and encrypted data **118** and send the compressed and decrypted data **116** to the client **102**. The client **102** may decompress the compressed and decrypted data **116** to restore the original data **114** which may be, for example, an object.

(56) FIG. 1B illustrates an embodiment of an object storage scheme with client-side encryption in accordance with example embodiments of the disclosure. The left side of FIG. 1B illustrates data flow between components of a system during read and/or write operations, and the right side of FIG. 1B illustrates operations on data during a write operation.

(57) The system illustrated on the left side of FIG. 1B and the operations illustrated on the right side of FIG. 1B may include some components and/or operations that may be similar to those illustrated in FIG. 1A and may be indicated by the same or similar reference numerals. However, in the embodiment illustrated in FIG. 1B, the client **102** may encrypt the compressed data **116** to generate compressed and encrypted data **118** as shown by the first group **110B** of operations performed by the client **102**. The client **102** may send the compressed and encrypted data **118** to the server **104** which may divide the compressed and encrypted data **118** into one or more data chunks **120** as shown by the second group of operations **112B** performed by the server **104**. The server **104** may send the one or more data chunks **120** to one or more storage devices **108**. In some embodiments, the server **104** may perform erasure coding on the one or more data chunks **120** to generate one or more parity chunks **121** which may also be stored on the one or more storage devices **108**.

(58) During a read operation, the operations shown in FIG. 1B may be performed in reverse. For example, the server **104** may reconstruct the compressed and encrypted data **118** from the data chunks **120** (recovering any missing or corrupted data chunk using the one or more parity chunks **121** if needed) and send the compressed and encrypted data **118** to the client **102**. The client **102** may decrypt the compressed and encrypted data **118** to generate the compressed and decrypted data **116**. The client **102** may decompress the compressed and decrypted data **116** to restore the original data **114** which may be, for example, an object.

(59) The embodiments illustrated in FIG. 1A and FIG. 1B are example embodiments only, and the number, order, and/or arrangement of components and/or operations may be varied. For example, in some implementations, the original data **114** may be stored without compression and/or without encryption. In some embodiments, the one or more servers **104** may be implemented with a first server that may be configured as an object storage server and a second server that may be configured as a storage server (which may also be referred to as a storage node) to manage the one or more storage devices **108**. Thus, the first and second servers may implement an object storage service. If any or all of the original data **114** is encrypted, encryption keys may be generated by the storage service and/or by a user of the service. In some embodiments, performing the chunking operation at or near the end of a write operation may enable the server **104** to divide the data into chunks having sizes that may correspond to one or more blocks sizes of the one or more storage devices **108**.

(60) In some situations, a user in association with a user's device may only need to retrieve a subset of data stored in an object. Some object storage systems may require the user to retrieve the entire object and process the object to find the subset of data. This may result in relatively large amounts of unneeded data being transmitted to the user's device, which in turn, may consume unnecessary resources such as time, bandwidth, power, and/or the like.

(61) To reduce and/or prevent the transmission of unneeded data, some object storage systems may provide a data selection feature that may enable a user to request a specified subset of data to retrieve from a stored object. Rather than sending the entire object to the user's device, the object storage system may perform a scanning, filtering, and/or other data selection operation on the object to find the specified subset of data. The object storage system may return the specified subset of data to the user's device.

(62) FIG. 2A illustrates an embodiment of an object storage scheme that may return an object to a user's device in accordance with example embodiments of the disclosure. FIG. 2B illustrates an embodiment of an object storage scheme having a data selection feature in accordance with example embodiments of the disclosure.

(63) Referring to FIG. 2A, an object storage service **201** may store objects **203A**, **203B**, and **203C** for a user in a data bucket or container **205**. If the user needs to retrieve a subset of data (e.g., one or more records) from one of the objects **203A**, the object storage service **201** may require the user to request the entire object **203A** which may be sent to a client compute operation **207** over a network. The client compute operation **207** may perform a data selection operation **209** such as scanning, filtering, and/or the like, on the object **203A** to find the subset of data. The client compute operation **207** may use the subset of data for a further operation **211**.

(64) Referring to FIG. 2B, an object storage service **213** having a data selection feature may enable a user to request a subset of data from a stored object **203A**. For example, the object storage service **213** may enable a user to submit a request, for example, by sending a query (e.g., an expression using a database language such as SQL) that may operate on the object **203A** which may be stored, for example, in a format such as comma separated variables (CSV), JavaScript Object Notation (JSON), Parquet, and/or the like. In some embodiments, the query may be sent to the object storage service **213**, for example, using an application programming interface (API), software development kit (SDK), and/or the like.

(65) Rather than sending the entire object **203A**, the object storage service **213** may perform a data selection operation **209** such as scanning, filtering, and/or the like on the object **203A** to find the subset of data specified by the user in the request. The object storage service **213** may send the subset of data **213a** to a client compute operation **217** for a further operation **211**. Depending on the implementation details, the object storage service **213** may perform one or more restore operations **219** on the object **203A** such as decompression, decryption, and/or the like, to reverse a compression operation, encryption operation, and/or the like that may have been performed on the object **203A** when it was stored.

(66) FIG. 3A illustrates an embodiment of a write operation of an object storage scheme having a data selection feature in accordance with example embodiments of the disclosure. FIG. 3B illustrates an embodiment of a read operation of an object storage scheme having a data selection feature in accordance with example embodiments of the disclosure. The embodiments illustrated in FIG. 3A and FIG. 3B may be used, for example, to implement the object storage scheme illustrated in FIG. 2B.

(67) The left side of FIG. 3A illustrates data flow between components of an object storage system during read and/or write operations, and the right side of FIG. 3A illustrates operations on data during a write operation.

(68) The system illustrated on the left side of FIG. 3A may include a client **302**, one or more servers **304** (which may be referred to collectively as a server), and one or more storage devices **308** (which may be referred to collectively as storage). The operations illustrated on the right side of FIG. 3A are shown in a first group **310A** performed by the client **302** and a second group **312A** performed by the server **304**. The data flow between components and/or operations on data illustrated in FIG. 3A may be similar to the embodiment with server-side encryption illustrated in FIG. 1A or the embodiment with client-side encryption illustrated in FIG. 1B in which elements having reference numerals ending in the same digits may be similar. Thus, in FIG. 3A, the compressed and encrypted data **318** may be part of group **310A** for an implementation with client-side encryption, or part of group **312A** for an implementation with server-side encryption.

(69) Referring to FIG. 3B, a user may request a subset of data from an object or other original data stored on the one or more storage devices **308**. To process such a request, the server **304** may read one or more data chunks **320** from the one or more storage devices **308**. If one of the data chunks is missing or corrupted, the server **304** may recover the missing or corrupted data chunk using the one or more parity chunks **321**. The server **304** may reconstruct the compressed and encrypted data **318** from the data chunks **320**.

(70) The server **304** may decrypt the compressed and encrypted data **318** to generate the compressed and decrypted data **316**, which may be decompressed to restore the original data **314**.

(e.g., an object). The server **304** may perform a data selection operation (e.g., scanning, filtering, and/or the like) on the original data **314** to obtain the requested subset of data **323**. The server **304** may send the subset of data **323** to the client **302**. Because the decompression operation of the client may be bypassed, it is grayed-out. The operations illustrated on the right side of FIG. **3B** are shown in a group **312B** performed by the server **304**.

(71) As with the embodiments illustrated in FIG. **1A** and FIG. **1B**, the server **304** illustrated in FIG. **3A** and FIG. **3B** may be implemented with a first server that may be configured as an object storage server and a second server that may be configured as a storage server to manage the one or more storage devices **308**. Thus, in some embodiments, a storage server may reconstruct the compressed and encrypted data **318** from the one or more data chunks **320**, and an object storage server may perform the decryption, decompression, and/or data selection operations. Moreover, although the embodiments illustrated in FIG. **3A** and FIG. **3B** may implement server-side encryption, other embodiments may implement client-side encryption.

(72) Depending on the implementation details, the embodiments illustrated in FIG. **3A** and FIG. **3B** may reduce network traffic, for example, by reducing the amount of data transferred between a storage system and a client. However, the data processing flow for the architecture illustrated in FIG. **3A** and FIG. **3B** may prevent the storage system from taking advantage of computational storage devices which, depending on the implementation details, may be well-suited to performing some or all of the operations performed by the server **304**. For example, in some embodiments, a computational storage device may include processing resources that may perform decompression, decryption, and/or other operations such as data selection operations more efficiently than the general purpose processing resources that may be present in a server. However, because the original data **314** may be modified (e.g., compressed, encrypted, and/or the like) prior to chunking, an individual storage device **308** on which a data chunk is stored may not be able to decrypt, decompress, and/or otherwise restore the chunk of data to a form on which a meaningful operation may be performed locally at the device.

(73) FIG. **4** illustrates an embodiment of a storage system having local data restoration in accordance with example embodiments of the disclosure. The system illustrated in FIG. **4** may include a host **424** and a computational storage device **408** that may communicate through a connection **422**. The host **424** may include data chunking logic **426** (which may also be referred to as data dividing logic) and data modification logic **427** that may be configured to provide one or more chunks of data to the storage device **408** in a form in which the storage device **408** may restore a chunk of data to a form on which the storage device may perform an operation. For example, the data chunking logic **426** may divide an object or other original data into one or more chunks of data prior to modification by the data modification logic **427**. The data chunking logic **426** may or may not be content aware. For example, in an embodiment in which the data chunking logic **426** is not content aware, the data chunking logic **426** may divide data into fixed sized chunks, and thus, one or more chunks may include one or more fragments of records. As another example, in an embodiment in which the data chunking logic **426** is content aware, the data chunking logic **426** may look for a record delimiter in the data to divide the data into variable sized chunks such that a chunk may end with a record delimiter, and therefore, end with a complete record.

(74) The data modification logic **427** may perform one or more data modification operations such as compression, encryption, erasure coding, and/or the like, on one or more of the chunks individually to generate one or more modified chunks of the original data. The host **424** may send one or more of the modified chunks of the original data to the computational storage device **408** and/or to one or more additional computational storage devices for storage and/or processing.

(75) The computational storage device **408** may include data restoration logic **428**, one or more processing elements **429**, and storage media **430**. The data restoration logic **428** may be configured to restore a modified chunk of data to a form on which the one or more processing elements **429**

may perform an operation. For example, the data restoration logic **428** may decrypt a modified chunk of data if it was encrypted, decompress a modified chunk of data if it was compressed, and/or the like. The one or more processing elements **429** may be configured to perform any type of operation such as data selection (e.g., scanning, filtering, and/or the like), compute acceleration, graph processing, graphics processing, machine learning, and/or the like. The storage media **430** may be used to store any data including one or more modified chunks of data sent by the host **424**.

(76) In some embodiments, the data restoration logic **428** and/or one or more processing elements **429** may be configured to read and restore one or more chunks of data from the storage media **430** and return a specified subset of the data, or perform any other operation on the restored chunk of data, in response to a request which may include a query (e.g., an expression) received at the storage device **408**.

(77) In some embodiments, a restored chunk of data may or may not be the exact same as the original data prior to chunking. For example, if a chunk of data stored at the storage device **424** contains financial information such as bank account transactions, balances, and/or the like, and the user requests just the account balances, the restoration logic **428** and/or one or more processing elements **429** may need to restore the chunk of data to the original form to find the exact account balances and send them to the user's device. However, if a chunk of data stored at the storage device **424** contains a photographic image, and the user requests a list of features in the image, the restoration logic **428** and/or one or more processing elements **429** may only need to decompress the image to an extent that may enable the one or more processing elements **429** to identify the features requested by the user.

(78) The host **424** may be implemented with any component or combination of components that may provide one or more chunks of data to the storage device **408** in a form in which the storage device **408** may restore and/or perform an operation on. For example, in some embodiments, the host **424** may include a client, an object storage server, and/or a storage node. The data chunking logic **426** and/or data modification logic **427** may be distributed between any components of the host **424** in any manner. For example, in some embodiments, the data chunking logic **426** may be implemented at a client whereas the data modification logic **427** may be implemented at an object storage server and/or a storage node. As another example, the data chunking logic **426** and a portion of the data modification logic **427** including compression logic may be implemented at a client, whereas a portion of the data modification logic **427** including encryption and/or erasure coding logic may be implemented at a server. Thus, the client may divide original data into chunks, individually compress the chunks of data, and send the compressed chunks of data to the server. The server may individually encrypt the compressed chunks of data, perform erasure coding on the chunks of data to generate one or more parity chunks, and store the chunks of data and/or parity chunks over one or more storage devices including the computational storage device **408**.

(79) The storage device **408**, and/or any other storage devices disclosed herein, may be implemented in any form factor such as 3.5 inch, 2.5 inch, 1.8 inch, M.2, Enterprise and Data Center SSD Form Factor (EDSFF), NF1, and/or the like, using any connector configuration such as Serial ATA (SATA), Small Computer System Interface (SCSI), Serial Attached SCSI (SAS), M.2, U.2, U.3 and/or the like.

(80) The storage device **408**, and/or any other storage devices disclosed herein, may be implemented with any storage media **430** including solid state media, magnetic media, optical media, and/or the like, or any combination thereof. Examples of solid state media may include flash memory such as not-AND (NAND) flash memory, low-latency NAND flash memory, persistent memory (PMEM) such as cross-gridded nonvolatile memory, memory with bulk resistance change, phase change memory (PCM), and/or the like, or any combination thereof.

(81) The storage device **408**, and/or any other storage devices disclosed herein, may communicate using any type of storage interface and/or protocol such as Peripheral Component Interconnect Express (PCIe), NVMe, NVMe-over-fabric (NVMe-oF), NVMe Key-Value (NVMe-KV), SATA,

SCSI, and/or the like, or any combination thereof. In some embodiments, the storage device **408**, and/or any other storage devices disclosed herein, may implement a coherent (e.g., memory coherent, cache coherent, and/or the like) or memory semantic interface such as Compute Express Link (CXL), and/or a coherent protocol such as CXL.mem, CXL.cache, and/or CXL.IO. Other examples of coherent and/or memory semantic interfaces and/or protocols may include Gen-Z, Coherent Accelerator Processor Interface (CAPI), Cache Coherent Interconnect for Accelerators (CCIX), and/or the like.

(82) The storage device **408**, and/or any other storage devices disclosed herein, as well as any components of the host **424** (e.g., a client, an object storage server, a storage node, and/or the like) may be implemented entirely or partially with, and/or used in connection with, a server chassis, server rack, dataroom, datacenter, edge datacenter, mobile edge datacenter, and/or any combinations thereof.

(83) The communication connection **422**, and/or any other connections disclosed herein, including any connections between components such as clients, servers, storage devices, and/or the like, may be implemented with any interconnect and/or network interfaces and/or protocols including PCIe, Ethernet, Transmission Control Protocol/Internet Protocol (TCP/IP), remote direct memory access (RDMA), RDMA over Converged Ethernet (ROCE), FibreChannel, InfiniBand, iWARP, and/or the like, or any combination thereof.

(84) Any of the functionality disclosed herein, including any of the logic such as the data chunking logic **426**, data modification logic **427**, data restoration logic **428**, one or more processing elements, **429**, indication logic **531**, and/or the like, may be implemented with hardware, software or a combination thereof including combinational logic, sequential logic, one or more timers, counters, registers, and/or state machines, one or more complex programmable logic devices (CPLDs), field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), central processing units (CPUs) such as complex instruction set computer (CISC) processors such as x86 processors and/or reduced instruction set computer (RISC) processors such as ARM processors, graphics processing units (GPUs), neural processing units (NPU)s, tensor processing units (TPUs) and/or the like, executing instructions stored in any type of memory, or any combination thereof. In some embodiments, one or more of the data restoration logic **428**, processing elements **429**, and/or the like may include fixed and/or programmable functionality to perform any functions such as compression and/or decompression, encryption and/or decryption, microservices, erasure coding, video encoding and/or decoding, database acceleration, searching, machine learning, graph processing, and/or the like. In some embodiments, one or more components may be implemented as a system-on-chip (SOC).

(85) In some embodiments, one or more of the data restoration logic **428**, processing elements **429**, and/or the like may be integrated with one or more other components of a storage device such as a storage device controller, a flash translation layer (FTL) and/or the like.

(86) Any of the data modification operations disclosed herein such as compression, encryption, and/or the like (or reverse operations thereof), may be implemented with any suitable techniques. For example, data compression and/or decompression may be implemented with LZ77, gzip, Snappy, and/or the like. Encryption and/or decryption may be implemented with Advanced Encryption Standard (AES) such as AES-256, Rivest-Shamir-Adleman (RSA), and/or the like.

(87) FIG. 5 illustrates another embodiment of a storage system having local data restoration in accordance with example embodiments of the disclosure. The system illustrated in FIG. 5 may include components and/or implement operations similar to those described with respect to the embodiment illustrated in FIG. 4 in which elements having reference numerals ending in the same digits may be similar. However, in the embodiment illustrated in FIG. 5, the computational storage device **508** may further include indication logic **531** that may be configured to provide one or more indications **532** to the data chunking logic **526** and/or the data modification logic **527** at the host **524**.

(88) The one or more indications **532** may include information that may be used by the data chunking logic **526** to determine how to divide original data into chunks. For example, the one or more indications **532** may include one or more storage hyper-parameters such as a minimum chunk size, maximum chunk size, optimal chunk size, and/or the like for storage utilization, processing efficiency (e.g., chunk decompression, decrypting, data selection, and/or other operations), bandwidth utilization, and/or the like.

(89) The one or more indications **532** (e.g., storage hyper-parameters) may include information that may be used by the data modification logic **527** to determine how to modify the individual chunks of data provided by the data chunking logic **526**. For example, the one or more indications **532** may include a list of the types of compression algorithms, encryption algorithms, and/or the like, supported by the data restoration logic **528** at the storage device **508**.

(90) In some embodiments, one or more indications may be mandatory, optional (e.g., provided as a suggestion), or a combination thereof. For example, an indication of an optimal chunk size for storage on the storage device **508** may be provided as a suggestion, whereas an indication of one or more compression algorithms, encryption algorithms, and/or the like supported by the data restoration logic **528**, may be mandatory to enable the storage device **508** to decompress and/or decrypt a chunk of data for local processing by the one or more processing elements **529** at the storage device **508**.

(91) In some embodiments, the indication logic **531** may be located entirely at the computational storage device **508**. In some other embodiments, however, the indication logic **531** may be located at the host **524**, distributed between the host **524** and the storage device **508** or multiple storage devices, or located entirely at a different apparatus (e.g., a separate server, a management controller, and/or the like, that may maintain a list or database of characteristics of storage devices in a system). For example, in some embodiments, one or more storage nodes may include indication logic **531** that may maintain a list or database of indications for each storage device installed at the storage node and provide the indications to one or more clients, object storage servers, and/or the like. As a further example, one or more storage nodes may include a portion of indication logic **531** that may maintain indications for each storage device installed at the storage node, and an object storage server may include a portion of indication logic **531** that may aggregate indications from one or more storage nodes and provide the indications to one or more clients.

(92) Any of the indications **532** may be provided to any apparatus such as a client, an object storage server, a storage node, and/or the like, by the indication logic **531**, for example, in response to a query, a command, and/or the like (e.g., an NVMe command, a query through an API, an SDK, and/or the like). In some embodiments, the one or more indications **532** (e.g., one or more storage hyper-parameters) may be provided to a user's device (e.g., by a client) through a client library.

(93) FIG. **6A** illustrates an example embodiment of a write operation for a storage scheme having local data restoration and server-side encryption in accordance with example embodiments of the disclosure. The embodiment illustrated in FIG. **6A** may be used, for example, to implement any of the storage schemes illustrated in FIG. **4** and FIG. **5**. The left side of FIG. **6A** illustrates data flow between components of a storage system, and the right side of FIG. **3A** illustrates operations on data during the write operation.

(94) The system illustrated on the left side of FIG. **6A** may include a client **602**, one or more servers **604** (which may be referred to collectively as a server), and one or more storage devices **608** (which may be referred to collectively as storage). The operations illustrated on the right side of FIG. **6A** are shown in a first group **610A** performed by the client **602** and a second group **612A** performed by the server **604**.

(95) A write operation may begin when a storage device **608** and/or server **604** provide one or more indications **632** to the client **602** indicating a data chunk size, compression algorithm, and/or the like. The client **602** may divide original data **614** into one or more chunks **633** based, for example, on the one or more indications **632**.

(96) Referring back to FIG. 3A and FIG. 3B, the data chunks **320** may be essentially the same size which may be required, for example, when the storage **308** is implemented with one or more block-based storage devices. In the embodiment illustrated in FIG. 6A, however, the data chunks **633** may be different sizes, for example, to take advantage of one or more of the storage devices **608** that may be implemented with KV interfaces. Additionally, or alternatively, the server **604** may implement software emulation of a key-value interface (e.g., RocksDB, LevelDB, and/or the like) on top of one or more block-based storage devices **608**. Although the data chunks illustrated in FIG. 3A and FIG. 3B are shown with different sizes, the principles may also be applied to systems in which some or all of the storage devices have block-based interfaces, which may be considered a subset of variable sized chunks.

(97) In some embodiments, an advisory and/or mandatory chunk size may be determined, for example, based on a chunk size that may be the best known size for a specific storage device. For example, with some solid state drives (SSDs), a 128 KB chunk size may fully utilize the SSD bandwidth. Additionally, or alternatively, a storage server may provide an optimal chunk size to the client **602** through a library, and the client **602** may internally split an object or other original data into smaller chunks when the user stores the object or other original data. Additionally, or alternatively, the client **602** may analyze the content and dynamically determine the chunk size.

(98) After chunking the original data **614**, the client may individually compress one or more of the data chunks **633** to generate one or more compressed chunks **634**. The client **602** may send the compressed chunks **634** to the server **604** which may encrypt the one or more compressed chunks **634** to generate one or more compressed and encrypted data chunks **635**. The server **604** may perform erasure coding on the one or more compressed and encrypted data chunks **635** to generate one or more parity chunks **636** and store the one or more data chunks **635** and one or more parity chunks **636** across one or more storage devices **608**.

(99) FIG. 6B illustrates an example embodiment of a write operation for a storage scheme having local data restoration and client-side encryption in accordance with example embodiments of the disclosure. The embodiment illustrated in FIG. 6B may be used, for example, to implement any of the storage schemes illustrated in FIG. 4 and FIG. 5. The left side of FIG. 6B illustrates data flow between components of a storage system, and the right side of FIG. 6B illustrates operations on data during the write operation.

(100) The data flow between components and/or operations on data illustrated in FIG. 6B may be similar to the embodiment with server-side encryption illustrated in FIG. 6A, and elements having reference numerals ending in the same digits may be similar. However, in the embodiment illustrated in FIG. 6B, the client **602** may encrypt the one or more compressed chunks of data **634** to generate the one or more compressed and encrypted data chunks **635** as shown in the first operation group **610B**. The client **602** may send the one or more compressed and encrypted data chunks **635** to the server **604** which may perform erasure coding on the one or more compressed and encrypted data chunks **635** to generate one or more parity chunks **636** and store the one or more data chunks **635** and one or more parity chunks **636** across one or more storage devices **608** as shown in operation group **612B**.

(101) After the one or more chunks of data **633** have been individually modified (e.g., compressed, encrypted, and/or the like) and stored as modified data chunks **636** across one or more storage devices **608**, each storage device may be able to restore one or more data chunks (e.g., by decrypting and/or decompressing the one or more data chunks) and perform an operation on the restored data chunk. For example, a user, client **602**, server **604**, and/or the like, may send a request to one or more of the storage devices **608** to restore one or more of the chunks and perform one or more operations (e.g., a data selection operation) on the restored chunk of data.

(102) FIG. 7A illustrates an example embodiment of a write operation for a storage scheme having local data restoration in accordance with example embodiments of the disclosure. FIG. 7B illustrates an example embodiment of a read operation with data selection for a storage scheme

having local data restoration in accordance with example embodiments of the disclosure. The embodiments illustrated in FIG. 7A and FIG. 7B may be used, for example, to implement any of the storage schemes illustrated in FIG. 4 and FIG. 5.

(103) Referring to FIG. 7A, the left side illustrates data flow between components of a storage system, and the right side illustrates operations on data during the write operation. Referring to FIG. 7B, the left side illustrates data flow between components of a storage system, and the right side illustrates operations on data during the read operation.

(104) The write operation illustrated in FIG. 7A may implement server-side encryption similar to that illustrated in FIG. 6A or client-side encryption similar to that illustrated in FIG. 6B, and elements having reference numerals ending in the same digits may be similar. Thus, the data chunks **734**, which have been individually compressed, may be encrypted to generate the compressed and encrypted data chunks **735** as part of the client operations **710A** or part of the server operations **712A**.

(105) Referring to FIG. 7B, one or more computational storage devices **708** may receive one or more requests to perform a data selection operation to read one or more subsets of data from one or more chunks of data **735** stored at the storage device. The one or more requests may include, for example, one or more expressions to specify the requested subsets of data. The requests may be received, for example, from the client **702** through the server **704**.

(106) To process the one or more requests, the one or more storage devices **708** may perform a group of operations **737** locally at the one or more storage devices **708**. Each of three different storage devices may perform a group of data restoration and data selection operations **737-1**, **737-2**, and **737-3**, respectively, on a corresponding chunk of data stored at each device. However, in some embodiments, a single storage device may perform data restoration and data selection or other operations on any number of data chunks stored at the device.

(107) Each storage device **708** may read, from a storage media, a corresponding chunk of data **735** that has been individually compressed and encrypted. Each storage device may decrypt the corresponding chunk of data to generate a compressed and decrypted chunk of data **734**. Each storage device may decompress the corresponding chunk of data to generate a restored chunk of data **738**. In some embodiments, each restored chunk of data **738** may be identical to a corresponding portion of the original data **714**. However, in some embodiments, a restored chunk of data **738** may only be restored to a form that may enable the storage device **708** to perform a meaningful operation on the restored data (e.g., some embodiments may be able to perform one or more operations on a chunk of data that has not been completely decompressed).

(108) After the chunks of data have been restored, each storage device **708** may perform a data selection operation (e.g., scanning, filtering, and/or the like) based, for example, on an expression provided with the request, to obtain one or more corresponding results **739**. The one or more storage devices **708** may send the results **739** to the client as the one or more requested subsets **740** of the original data **714**. Because the decompression and/or decryption operations of the client may be bypassed, they are grayed-out.

(109) In some embodiments, one or more of the storage devices **708** may be able to recover one or more missing data chunks **735** if a parity chunk **736** is stored at the storage device. Alternatively, or additionally, a server **704** may restore one or more missing data chunks **735** using one or more parity chunks **736** stored at one or more other storage devices.

(110) Depending on the implementation details, performing a data recovery and/or a data selection operation at a storage device may reduce the time, bandwidth, power, latency, and/or the like, associated with reading a subset of original data (e.g., a subset of an object) stored in one or more chunks across one or more storage devices.

(111) FIG. 8 illustrates an example embodiment of a system architecture for an object storage scheme with local data restoration in accordance with example embodiments of the disclosure. The system illustrated in FIG. 8 may be used, for example, to implement any of the schemes described

with respect to FIG. 4, FIG. 5, FIG. 6A, FIG. 6B, FIG. 7A, FIG. 7B, FIG. 9A, and/or FIG. 9B.

(112) The system illustrated in FIG. 8 may include a client **802** and an object storage server cluster **804** connected through a network connection **842**. The system may also include one or more storage nodes **806** connected to the object storage server cluster **804** through a storage network **844**.

(113) The client **802** may include data chunking logic **826** and/or compression logic **846** which may be configured to perform data chunking of original data (e.g., one or more objects) prior to compressing individual chunks of data so the one or more computational storage devices **808** may restore a chunk of data to perform an operation on the restored chunk of data.

(114) The object storage server cluster **804** may include encryption logic **847**, erasure coding logic **848**, data selection logic **849**, cluster management logic **850**, and/or node and storage device management logic **851**. The encryption logic **847** may be used to individually encrypt chunks of data (e.g., compressed data) received from the client **802**. The erasure coding logic **848** may perform erasure coding of data chunks across storage nodes **806** and/or the storage devices **808**. The data selection logic **849** may perform various operations related to data restoration, data selection, and/or other processing operations performed by the individual storage devices **808**. For example, the data selection logic **849** may receive requests from the client **802** to read one or more subsets of data that may be stored in chunks across one or more storage devices **808**. The data selection logic **849** may forward the requests to the corresponding storage nodes **806** and/or storage devices **808**, receive and/or aggregate results from the corresponding storage nodes **806** and/or storage devices **808**, and send the aggregated results to the client **802**. The cluster management logic **850** may perform housekeeping and/or management functions related to maintaining the storage server cluster **804**. The node and storage device management logic **851** may perform housekeeping and/or management functions related to maintaining the one or more storage nodes **806** and/or storage devices **808**.

(115) Each of the storage nodes **806** may include a processing unit (e.g., a data processing unit (DPU), CPU, and/or the like) **852** and one or more computational storage devices **808**. The DPU **852** may perform various functions such as receiving and distributing requests from the client **802** to read one or more subsets of data that may be stored in chunks across one or more storage devices **808**. In some embodiments, the DPU **852** may perform data compression, data encryption, erasure coding, and/or the like, on chunks of data received from the object storage server cluster **804** and stored on the one or more computational storage devices **808**. In some embodiments, the DPU **852** may aggregate results of one or more data selection operations performed by the one or more computational storage devices **808** and forward the aggregated results to the object storage server cluster **804** and/or client **802**.

(116) Computational storage device **808a** shows an example of components that may be included in one or more of the computational storage devices **808**. The computational storage device **808a** may include a data selection engine **853** and storage media **830**. The data selection engine **853** may include decryption logic **854** and decompression logic **855** that may be used to decrypt and/or decompress chunks of data, respectively, that have been individually encrypted and/or compressed to restore the chunks of data to a form that may be operated on. The data selection engine **853** may also include data selection logic **856** that may be used to perform a data selection or other operation on a restored chunk of data. The data selection engine **853** may also include KV logic **857** that may be used to implement a KV interface for the storage device **808a**.

(117) In some embodiments, the system illustrated in FIG. 8 may be implemented with KV interfaces for some or all of the storage devices **808**. Depending on the implementation details, this may facilitate and/or enable the chunks of data to be implemented with variable chunk sizes. For purposes of illustration, the embodiment illustrated in FIG. 8 may be described as implementing a data selection feature with restored data chunks locally at one or more of the storage devices **808**, however, the principles may be applied to any type of processing that may be performed on

restored data chunks.

(118) FIG. 9A illustrates example embodiments of read and/or write operations for a storage scheme with local data restoration in accordance with example embodiments of the disclosure. The operations illustrated in FIG. 9A may be implemented, for example, using the system illustrated in FIG. 8. For purposes of illustration, a first group of operations **958A** may be assumed to be performed by the client **802** and object storage server cluster **804** illustrated in FIG. 8, and a second group of operations **959A** may be assumed to be performed by the one or more storage nodes **806** and/or storage devices **808** illustrated in FIG. 8, however, in other embodiments, the operations illustrated in FIG. 9A may be performed by any other components.

(119) Referring to FIG. 9A, during a write operation (e.g., a put operation) original data **914** (e.g., one or more objects) may be chunked by a client to generate one or more chunks of data **933**. The one or more chunks **933** may be individually compressed by the client to generate one or more compressed chunks **934**, which may be sent to, and encrypted individually by, an object storage server to generate one or more compressed and/or encrypted chunks **935**. The object storage server may perform erasure coding on the one or more compressed and/or encrypted chunks **935** to generate one or more parity chunks **936**.

(120) The object storage server may send the one or more compressed and encrypted chunks **935** and one or more parity chunks **936** (e.g., through a put operation **960**) to one or more storage nodes for storage over one or more storage devices. Thus, after the write operation, the original data **914** (e.g., an object) may be stored across one or more storage devices in one or more chunks **935** that may have been individually modified (e.g., compressed and/or encrypted).

(121) During a read operation (e.g., a get operation), for example, in an implementation in which a storage device may not recover and/or perform an operation on a chunk of data, one or more chunks of individually modified data **935** may be read from one or more storage devices. If one or more of the data chunks **935** is missing or corrupted, the missing and/or corrupted chunks may be recovered (e.g., by a storage device and/or a storage node) using the one or more parity chunks **936**.

(122) The one or more compressed and/or encrypted chunks **935** may be sent to an object storage server (e.g., through a get operation **962**) that may decrypt the one or more compressed and/or encrypted chunks **935** to generate one or more compressed and decrypted chunks **934**. The one or more compressed and decrypted chunks **934** may be sent to a client that may decompress the one or more data chunks **934** to generate decrypted and decompressed data chunks **933**, and assemble them back into the original data **914**.

(123) FIG. 9B illustrates an example embodiment of a read operation for a storage scheme with local data restoration and a data selection operation in accordance with example embodiments of the disclosure. The operations illustrated in FIG. 9B may be implemented, for example, using the system illustrated in FIG. 8. For purposes of illustration, a first group of operations **958B** may be assumed to be performed by the client **802** and/or object storage server cluster **804** illustrated in FIG. 8, and a second group of operations **959B** may be assumed to be performed by the one or more storage nodes **806** and/or storage devices **808** illustrated in FIG. 8, however, in other embodiments, the operations illustrated in FIG. 9B may be performed by any other components.

(124) To begin a read operation (e.g., a get operation **963**), one or more computational storage devices may receive one or more requests to perform a data selection operation to read one or more subsets of data from one or more chunks of data **935** stored at the one or more storage devices. The one or more requests may include, for example, one or more expressions to specify the requested subsets of data.

(125) To service the one or more requests, one or more chunks of individually modified data **935** may be read from one or more storage devices. The one or more storage devices may individually decrypt the one or more chunks of data **935** to generate one or more chunks of compressed and decrypted data **934**. The one or more storage devices may individually decompress the one or more

chunks of compressed and decrypted data **934** to generate one or more chunks of restored data **938**. In some embodiments, each restored chunk of data **938** may be identical to a corresponding portion of the original data **914**. However, in some embodiments, a restored chunk of data **938** may only be restored to a form that may enable the storage device to perform a meaningful operation on the restored data (e.g., some embodiments may be able to perform one or more operations on a chunk of data that has not been completely decompressed).

(126) The storage device may perform a data selection operation (e.g., scanning, filtering, and/or the like) on the one or more chunks of restored data **938** to find the one or more subsets of data **939** (indicated as results R) specified by the one or more requests. If a storage device has restored and performed a data selection operation on more than one chunk of data, the storage device may aggregate the results of the data selection operation to generate an aggregated result **940** which may be sent to an object storage server and to the client that sent the request. Additionally, or alternatively, the results R (e.g., subsets of data) **939** found by the data selection operations by multiple storage devices may be aggregated by a storage node and sent to an object storage server and to the client that sent the request.

(127) Table 1 illustrates some example data that may be stored in a storage system in accordance with example embodiments of the disclosure. For purposes of illustration, the data shown in Table 1 is for real estate listings, but the principles may be applied to any type of data. Each row of Table 1 may correspond to a record having seven entries: a record index, living space in square feet, number of bedrooms, number of bathrooms, zip code, year built, and list price. Thus, for example, the first eight records may be identified by indexes 1-8, respectively.

(128) TABLE-US-00001 TABLE 1 Living List Space Zip Year Price Index (sq ft) Bedrooms
Bathrooms Code Built (\$) 1 2222 3 3.5 32312 1981 250000 2 1628 3 2 32308 2009 185000 3 3824
5 4 32312 1954 399000 4 1137 3 2 32309 1993 150000 5 3560 6 4 32309 1973 315000 6 2893 4 3
32312 1994 699000 7 3631 4 3 32309 1996 649000 8 2483 4 3 32312 2016 399000 9 2100 5 3
32305 1926 497000 10 . . .

(129) FIG. **10** illustrates an embodiment of a distribution of the data from Table 1 across three data chunks at three computational storage devices in accordance with example embodiments of the disclosure. In the embodiment illustrated in FIG. **10**, a semicolon is used as a delimiter between the individual records (which may correspond to the rows shown in Table 1), but in other embodiments, other delimiting techniques may be used.

(130) Referring to FIG. **10**, the first two records (identified by indexes 1 and 2) may fit entirely within a first data chunk **1064A** stored on a first storage device **1008A**. The third record (identified by index 3 and indicated by entries with single underlining) may be split (e.g., fragmented) between data chunks **1064A** and **1064B** stored on the first and second storage devices **1008A** and **1008B**, respectively. The fourth and fifth records (identified by indexes 4 and 5) may fit entirely within the second data chunk **1064B** stored on the second storage device **1008B**. The sixth record (identified by index 6 and indicated by entries with single underlining) may be split between data chunks **1064B** and **1064C** stored on the second and third storage devices **1008B** and **1008C**. The seventh and eighth records (identified by indexes 7 and 8) may fit entirely within the third data chunk **1064C** stored on the third storage device **1008C**. The ninth record (identified by index 9 and indicated by entries with single underlining) may be split between the third **1064C** stored on the third storage device **1008C** and another chunk on another storage device.

(131) For purposes of illustration, the computational storage devices **1008A**, **1008B**, and **1008C** are shown as being implemented with data restoration logic and/or processing elements as described above that may enable the storage devices to restore an individually modified chunk of data **1035**, for example, by decryption (to generate a decrypted chunk of data **1034**) and/or decompression to generate a restored chunk of data **1038**, and perform an operation such as a data selection operation on the restored chunk of data **1038** to obtain a specified subset of data **1039** from one or more of the records in the data chunk stored on the device. However, the principles are not limited to these

implementation details and may be applied to any type of operation that may be performed on any type of data chunks stored on any type of computational storage devices. For purposes of illustration, some embodiments described herein may implement fixed size data chunks (e.g., as may be used with block-based storage devices), however, the principles may also be applied to embodiments that may implement variable size data chunks (e.g., as may be used with KV storage devices).

(132) In some embodiments, a record may correspond to an object. In some embodiments described herein, a record (e.g., a JSON object) may be assumed to be smaller than a chunk which, depending on the implementation details, may ensure that an object may span no more than two chunks. In some embodiments, a delimiter can be implemented as a simple character such as a semicolon. For example, for CSV objects, a delimiter may be implemented as a carriage return. Additionally, or alternatively, one or more delimiters may be determined by a hierarchy. Thus, detecting a delimiter may be more complex than a simple comparison. For example, for JSON objects, a pair of curly braces (“{ . . . }”) may define the JSON object. Moreover, in some embodiments, JSON objects may have nested JSON arrays, so the outermost pair of curly braces may define a single record. Thus, the delimiter may be defined by the outermost right curly brace (“}”).

(133) Referring again to FIG. 10, records that fit entirely within one of the storage devices (e.g., records 1, 2, 4, 5, 7, and 8) may be processed by the corresponding storage device. For example, if a client issues a read request for a data selection operation to return a subset of the data stored in Table 1 (e.g., the client sends a read request with an expression to return all records (or a portion thereof) having a year built after 1980), records 1, 2, 4, 5, 7, and 8 may be processed directly by the corresponding storage device. However, records 3, 6, and 9 may not be processed locally at a storage device because they are fragmented between data chunks at two different storage devices.

(134) FIG. 11 illustrates an example embodiment of a storage system in which a server may reconstruct records split between data chunks at different storage devices in accordance with example embodiments of the disclosure. The embodiment illustrated in FIG. 11 may include an object storage server **1104**, two storage nodes **1106A** and **1106B** coupled to the object storage server **1104** through a storage network, and computational storage devices **1108A**, **1108B**, **1108C**, **1108D**, and **1108E** which may store data chunks including records similar to those illustrated in FIG. 10.

(135) Referring to FIG. 11, each of the storage devices **1108A** through **1108E** may send results **1165** of a data selection operation it may perform on any complete records in its corresponding data chunk **1164A** through **1164E**, respectively (either directly, or through the storage node at which it is located). However, because records 3, 6, and 9 may not be processed locally at a storage device, the object storage server **1104** may reconstruct the split records in one or more aggregate buffers **1166**. In some embodiments, each aggregate buffer **1166** may reconstruct the split record between the *i*th device and the (*i*+1)th device. For example, storage device **1108A** may send a first portion (which may also be referred to as a fragment) of record 3 (e.g., the index, living space, bedrooms, bathrooms, zip code, and year built) located in data chunk **1164A** to the object storage server **1104** to be aggregated in a first buffer **1166A** with a second portion of record 3 (list price) located in data chunk **1164B** and sent by storage device **1108B**. In some embodiments, the object storage server **1104** may include *N* aggregate buffers where *N* may be the number of storage devices coupled to the object storage server **1104**.

(136) The object storage server **1104** may perform a selection operation on the reconstructed records 3, 6, and 9 in the aggregate buffers **1166A**, **1166B**, and **1166C**, respectively, to generate results **1167**. Thus, between the results **1165** sent by the individual storage devices, and the results **1167** generated from the aggregate buffers **1166**, the object storage server **1104** may obtain all subsets of data specified by the request and return the subsets to the client.

(137) However, depending on the implementation details, each of the fragments of records sent

from the storage devices **1108** to the object storage server **1104** may consume time, bandwidth, and/or power, increase latency, reduce the utilization of processing resources, and/or the like, and/or may result in the object storage server **1104** becoming a potential bottleneck.

(138) FIG. **12** illustrates an embodiment of a content-aware method for dividing data into chunks and compressing data in accordance with example embodiments of the disclosure. In the method illustrated in FIG. **12**, uncompressed input data **1268** (which may be stored in a buffer) shown at the top of FIG. **12** may include one or more records that may each terminate with a delimiter which, in this example, may be a semicolon. In other embodiments, however, other delimiters such as brackets, carriage return-line feed (CR-LF) sequences, and/or the like, may be used. In some embodiments, one or more records may be included in an array (e.g., a JSON array), a document (e.g., a JSON document), and/or the like within the input data **1268**. If a record ends at the end of an array, a document, and/or the like, a different delimiter may be used to indicate the end of the array, document, and/or the like, but the method illustrated in FIG. **12** may also interpret the different delimiter as the end of a record.

(139) During a first pass (Operation 1), the method may scan the input data **1268** and use it to generate a chunk of data until it reaches a default chunk size N . If the last data element in the chunk **1269** is a delimiter, the method may proceed to a second pass (Operation 2). If the last data element in the chunk is not a delimiter, the method may modify the size of the chunk so the end of the chunk aligns with the end of a delimiter. For example, the method may begin at position $N+1$ in a buffer holding the data **1268** and scan backward until it finds the delimiter **1270**. The method may reduce the size of the chunk (e.g., shrink the chunk) until the delimiter **1270** is the last data element in the chunk. Thus, the chunk may be reduced by a size equal to the shrink space S so the chunk may only include one or more complete records.

(140) Alternatively, the method may begin at position $N+1$ in a buffer holding the data **1268** and scan forward until it finds the delimiter **1271**. The method may extend the size of the chunk until the delimiter **1271** is the last data element in the chunk **1269** as shown in the middle of FIG. **12**. Thus, the chunk may be extended by a size equal to the extended space E so the chunk may only include one or more complete records.

(141) Once the chunk size is determined, the method may proceed to a second pass (Operation 2) in which the self-contained chunk (e.g., chunk **1269**), which may end with a delimiter and only include complete records, may be compressed to a length C using one or more compression algorithms to generate compressed data **1272**.

(142) Because the size of a chunk of uncompressed data may be reduced or extended, the size of the chunk of uncompressed data **1269** may be variable, and thus, the size of the chunk of compressed data **1272** may be variable.

(143) Whether the method reduces or extends the length of the chunk may depend on various factors such as a maximum and/or optimal data size for an object or key-value storage device the chunk may be sent to, or a block size for a block-based storage device the chunk may be sent to. In the case of an object or key-value storage device, the resulting chunk of compressed, self-contained data **1272** may be stored without further modification and/or processing.

(144) Depending on the implementation details, the content-aware data chunking method illustrated in FIG. **12** may improve the operation of a storage system such as that shown in FIG. **11**, for example, because fewer or no records may be split between chunks on different storage devices, and thus, fewer or no fragments of records may be transferred between storage devices, storage nodes, storage servers, and/or the like, to enable components to restore chunks of data having complete records and/or perform operations on complete records.

(145) The compression method illustrated in FIG. **12** may be implemented, for example, using any of the systems illustrated in FIG. **4**, FIG. **5**, and/or FIG. **9**, in which the chunking logic **426**, **526**, and/or **826** may include content-aware logic configured to determine a chunk size based on finding boundaries (e.g., delimiters) of records or other units of data in the uncompressed input data **1268**.

(146) Although the method illustrated in FIG. 12 may improve the operation of a storage system, in some embodiments, and depending on the implementation details, the overhead associated with the use of two passes may increase the processing burden, especially when scanning relatively large objects.

(147) Some additional content-aware data chunking techniques in accordance with example embodiments of the disclosure may integrate a chunking operation with another operation such as a data compression operation that may scan the data to be chunked. Depending on the implementation details, this may improve the efficiency of the chunking operation because it may exploit a scanning operation that was already being performed for purposes of compression. Thus, in some embodiments, it may reduce or eliminate overhead associated with two scanning passes.

(148) FIG. 13 illustrates an embodiment of a method for compressing data in accordance with example embodiments of the disclosure. FIG. 14 illustrates an embodiment of operations that may be performed by the method illustrated in FIG. 13. The method illustrated in FIG. 13 and FIG. 14 may be based, for example, on a compression algorithm such as LZ77. As illustrated at the top of FIG. 13, the method may operate on a stream of input data that may pass through a lookahead buffer 1374 and a window 1373 (which may also be implemented as a buffer). A portion (which may also be referred to as a part) of the stream of input data ahead of the lookahead buffer 1374 may be indicated as an invisible part 1375 because it may not be to the logic performing the compression. Depending on how much data has passed through the lookahead buffer 1374 and window 1373, there may be another part of invisible data after the window 1373.

(149) The method illustrated in FIG. 13 may perform the following operations as illustrated in FIG. 14. The embodiments illustrated in FIG. 13 and FIG. 14 may be implemented with data elements of any size, configuration, and/or the like, such as bytes, words, nibbles, and/or the like.

(150) (1) At a first operation 1477-1, the method may set the coding position to the first data element of the stream of input data as shown at the top of FIG. 13.

(151) (2) At a second operation 1477-2, the method may scan the stream of input data to look for a longest match of length L between one or more data elements (which may be referred to as literals) in the lookahead buffer 1374, and one or more data elements in the window 1373 as shown at the top of FIG. 13.

(152) (3) At a third operation 1477-3, if a match was found at the second operation (2) 1477-2, the method may output a pointer P which may include, for example, an offset from the coding position indicating the beginning location of the matched data in the window 1373 and the length L indicating the number of data elements that match as shown at the top of FIG. 13. The method may move the coding position and window 1373 forward by L data elements as shown at the bottom of FIG. 13. The method may proceed to a fifth operation (5) 1477-5. Depending on the length L, this may cause all or part of the invisible part 1375 of the input data stream to enter the lookahead buffer 1374 and a part 1375 of data length L previously in the window 1373 to become invisible (e.g., inaccessible to the logic performing the method).

(153) (4) At a fourth operation 1477-4, if a match was not found at the second operation (2) 1477-2, the method may output a null pointer (e.g., a pointer in which the offset and/or length L are zero) and the next data element (literal) in the lookahead buffer 1374. The method may move the coding position and the window 1373 forward by one data element.

(154) (5) at a fifth operation 1477-5, if the lookahead buffer 1374 is not empty, the method may return to the second operation (2) 1477-2. Otherwise, the method may terminate.

(155) Thus, the compression method illustrated in FIG. 13 and FIG. 14 may operate by checking if a sequence of one or more literals in the lookahead buffer 1374 has ever appeared within the range of the window 1373 and replacing it with a pointer (e.g., an index) an instance of that sequence in the window 1373. The stream of output data generated by the compression operation may be a series of literal data elements (e.g., bytes) and pointers that may indicate a sequence of data elements that is already in the output.

(156) In some embodiments, a record may include any complete data structure (e.g., a row in a database) that may be used and/or operated on as a unit. In some embodiments, a delimiter may be implemented with anything that may indicate a boundary of a record, e.g., one or more data elements (a semicolon, a bracket, a carriage return-line feed (CR-LF) sequence, and/or the like). In some embodiments, a delimiter may be implemented with a data structure such as an index table that may indicate where a record ends.

(157) Because the compression method illustrated in FIG. 13 and FIG. 14 may scan input data as part of its normal operation, a chunking operation in accordance with example embodiments of the disclosure may be integrated into the compression method illustrated in FIG. 13 and FIG. 14 to exploit the data scanning already being performed by the compression operation.

(158) FIG. 15 illustrates an embodiment of an integrated chunking and compression scheme in accordance with example embodiments of the disclosure.

(159) As illustrated at the top of FIG. 15, a stream of input data may pass through a lookahead buffer 1574 and a window 1573 (which may also be implemented as a buffer). A part of the stream of input data ahead of the lookahead buffer 1574 may be indicated as an invisible part 1575 because it may not be to the logic performing the compression. Depending on how much data has passed through the lookahead buffer 1574 and window 1573, there may be another part of invisible data after the window 1573.

(160) FIG. 16 illustrates an embodiment of an integrated chunking and compression method that may be performed with the scheme illustrated in FIG. 15 in accordance with example embodiments of the disclosure.

(161) The embodiments illustrated in FIG. 15 and/or FIG. 16 may be implemented with data elements of any size, configuration, and/or the like, such as bytes, words, nibbles, and/or the like. The embodiments illustrated in FIG. 15 and/or FIG. 16 may use a current chunk size (c), an early termination indicator (e.g., an early termination bit) (e), and a record delimiter length (l). In some embodiments, the scheme illustrated in FIG. 15 may perform the following operations as illustrated in FIG. 16.

(162) (1) At a first operation 1678-1, the method may set the coding position to the first data element of the stream of input data as shown at the top of FIG. 15. In some embodiments, the current chunk size (c), an early termination bit (e), and/or a record delimiter length (l) may be initialized to zero. In some embodiments, the initial coding position may correspond to the start of a chunk of data as shown in the middle and at the bottom of FIG. 15.

(163) (2) At a second operation 1678-2, the method may scan the stream of input data to look for a longest match of length L between one or more data elements in the lookahead buffer 1574, and one or more data elements in the window 1573 as shown at the top of FIG. 15. While scanning the stream of input data to look for a longest match, the method may also look for a match for a delimiter for a record. If a match for a delimiter is found, the method may keep (e.g., record the position of) the first instance of the delimiter match, for example, in the window 1573 and set the delimiter length l to the length of the delimiter match that was found.

(164) (3) At a third operation 1678-3, if a longest match was not found at the second operation (2) 1678-2, the method may output a null pointer (e.g., a pointer in which the offset and/or length L are zero) and the next data element (literal) in the lookahead buffer 1574 to generate a stream of compressed output data. The method may move the coding position and the window 1573 forward by one data element. The method may also set the length L to one which may indicate that the current chunk size will be increased by the one data element output from the lookahead buffer 1574.

(165) (4) At a fourth operation 1678-4, if the early termination bit (e) is not set, and the current chunk size (c+L) is greater than or equal to a default chunk size (N), the method may set the early termination bit to one (e=1).

(166) (5) At a fifth operation 1678-5, if the early termination bit is set (e=1) and a delimiter match

has been found (e.g., $l \neq 0$), the method may set L equal to l (which is greater than one) and update the pointer P with the first matched delimiter (e.g., with the offset from the coding position indicating the beginning location of the first matched delimiter in the window **1573**, and the length l indicating the length of the matched delimiter).

(167) (6) At a sixth operation **1678-6**, if a match is found (e.g., a longest match or a delimiter match) the method may output the pointer P , for example, based on the location and length of the longest match (e.g., a normal match) as shown in the middle of FIG. **15** or the location and length of the second delimiter match (e.g., early termination) as shown at the bottom of FIG. **15** to generate the stream of compressed output data.

(168) (7) At a seventh operation **1678-7**, the method may move the coding position and window **1573** forward by L data elements, where L may be the length of a longest match as shown in the normal operation in the middle of FIG. **15** or L may be the length l of a delimiter match as shown in the early termination operation at the bottom of FIG. **15**. After the seventh operation, the stream of data may have one or more invisible parts **1575A**, **1575B**, and/or **1575C** depending on the lengths of L and l .

(169) (8) At an eighth operation **1678-8**, if the lookahead buffer **1574** is not empty, the method may return to the second operation (2) **1678-2**. Otherwise, the end of the chunk may be determined by the end of the delimiter as shown at the coding position at the bottom of FIG. **15**.

(170) In some embodiments, the scheme illustrated in FIG. **15** and/or the method illustrated in FIG. **16**, may initially perform a compression operation by finding longest matches similar to the method illustrated in FIG. **13** and FIG. **14** until the default chunk size N is reached. However, when approaching the default chunk size, the embodiments illustrated in FIG. **15** and/or FIG. **16** may compress the input data record-by-record until it exceeds the default chunk size. For example, in some embodiments, the method may find the smallest chunk that is larger than the default chunk size (and which may be terminated with delimiters). In some embodiments, this may be referred to as a minimum largest chunk. As the method scans the stream of input data, the start of the chunk as shown in the middle and at the bottom of FIG. **15** may progressively move away from the coding position.

(171) Depending on the implementation details, the content-aware data chunking techniques illustrated in FIG. **15** and/or FIG. **16** may improve the operation of a storage system such as that shown in FIG. **11**, for example, because fewer or no records may be split between chunks on different storage devices, and thus, fewer or no fragments of records may be transferred between storage devices, storage nodes, storage servers, and/or the like, to enable components to restore chunks of data having complete records and/or perform operations on complete records.

(172) Depending on the implementation details, the embodiments illustrated in FIG. **15** and/or FIG. **16** may also improve the efficiency of a data chunking operation because it may exploit a scanning operation that was already being performed for purposes of compression. Thus, in some embodiments, they may reduce or eliminate overhead associated with two scanning passes.

(173) FIG. **17** illustrates another embodiment of an integrated chunking and compression scheme in accordance with example embodiments of the disclosure.

(174) FIG. **18** illustrates an embodiment of an integrated chunking and compression method that may be performed with the scheme illustrated in FIG. **17** in accordance with example embodiments of the disclosure.

(175) The scheme illustrated in FIG. **17** may include components and/or implement operations similar to those described with respect to the embodiment illustrated in FIG. **15** in which elements having reference numerals ending in the same digits may be similar. However, in the embodiment illustrated in FIG. **17**, rather than compressing input data record-by-record, the method may determine the chunk size when the first record is encountered. For example, in some embodiments, the method may find the largest chunk that is smaller than the default chunk size (and which may be terminated with delimiters). In some embodiments, this may be referred to as a maximum

smallest chunk.

(176) As illustrated at the top of FIG. 17, a stream of input data may pass through a lookahead buffer 1774 and a window 1773 (which may also be implemented as a buffer). A part of the stream of input data ahead of the lookahead buffer 1774 may be indicated as an invisible part 1775 because it may not be to the logic performing the compression. Depending on how much data has passed through the lookahead buffer 1774 and window 1773, there may be another part of invisible data after the window 1773.

(177) The embodiment illustrated in FIG. 17 may perform one or more of the following operations as illustrated in FIG. 18. The embodiments illustrated in FIG. 17 and/or FIG. 18 may be implemented with data elements of any size, configuration, and/or the like, such as bytes, words, nibbles, and/or the like. The embodiments illustrated in FIG. 17 and/or FIG. 18 may use a current chunk size (c), an early termination indicator (e.g., an early termination bit) (e), and record delimiter length (l).

(178) (1) At a first operation 1879-1, the method may set the coding position to the first data element of the stream of input data as shown at the top of FIG. 17. In some embodiments, the current chunk size (c), an early termination bit (e), and/or record delimiter length (l) may be initialized to zero.

(179) (2) At a second operation 1879-2, the method may scan the stream of input data to look for a longest match of length L between one or more data elements in the lookahead buffer 1774, and one or more data elements in the window 1773 as shown at the top of FIG. 17. While scanning the stream of input data to look for a longest match, the method may also look for a match for a delimiter for a record. If a match for a delimiter is found, the method may keep the first instance of the delimiter match, for example, in the window 1773 and set the delimiter length l to the length of the delimiter match that was found.

(180) (3) At a third operation 1879-3, if a longest match was not found at the second operation (2) 1879-2, the method may output a null pointer (e.g., a pointer in which the offset and/or length L are zero) and the next data element (literal) in the lookahead buffer 1774 to generate a stream of compressed output data. The method may move the coding position and the window 1773 forward by one data element. The method may also set the length L to one which may indicate that the current chunk size will be increased by the one data element output from the lookahead buffer 1774.

(181) (4) At a fourth operation 1879-4, if the current chunk size (c+L) is greater than or equal to a default chunk size (N), the method may set the early termination bit to one (e=1).

(182) (5) At a fifth operation 1879-5, if the early termination bit is set (e=1) and a delimiter match has been found (e.g., l!=0), the method may set L equal to l (which is greater than one) and update the pointer P with the first matched delimiter (e.g., with the offset from the coding position indicating the beginning location of the first matched delimiter in the window 1773, and the length l indicating the length of the matched delimiter).

(183) (6) At a sixth operation 1879-6, if a match is found (e.g., a longest match or a delimiter match) the method may output the pointer P, for example, based on the location and length of the longest match (e.g., a normal match) as shown in the middle of FIG. 17 or the location and length of the delimiter match (e.g., early termination) as shown at the bottom of FIG. 17 to generate the stream of compressed output data.

(184) (7) At a seventh operation 1879-7, the method may move the coding position and window 1773 forward by L data elements, where L may be the length of a longest match as shown in the normal operation in the middle of FIG. 17 or L may be the length l of a delimiter match as shown in the early termination operation at the bottom of FIG. 17. After the seventh operation, the stream of data may have one or more invisible parts 1775A, 1775B, and/or 1775C depending on the lengths of L and l.

(185) (8) At an eighth operation 1878-8, if the early termination bit is not set (e!=1) and the

lookahead buffer **1774** is not empty, the method may return to the second operation (2) **1879-2**. Otherwise, the end of the chunk may be determined by the end of the delimiter as shown at the coding position at the bottom of FIG. **17**.

(186) In some embodiments, because the scheme illustrated in FIG. **17** and/or the method illustrated in FIG. **18** may determine the chunk size when the first record is encountered (maximum smallest chunk), the chunk size may be based on the first delimiter match as shown in the early termination operation at the bottom of FIG. **17**. As the method scans the stream of input data, the start of the chunk as shown in the middle and at the bottom of FIG. **17** may progressively move away from the coding position.

(187) Depending on the implementation details, the embodiments illustrated in FIG. **17** and/or FIG. **18** may improve the efficiency of a data chunking operation because it may exploit a scanning operation that was already being performed for purposes of compression. Thus, in some embodiments, it may reduce or eliminate overhead associated with two scanning passes.

(188) FIG. **19** illustrates another embodiment of an integrated chunking and compression method in accordance with example embodiments of the disclosure. The embodiment illustrated in FIG. **19** may be used, for example, with either or both of the embodiments illustrated in FIG. **15** and/or FIG. **17**. In some embodiments, the method illustrated in FIG. **19** may implement a minimum largest chunk and/or a maximum smallest chunk. For example, depending on the implementation details, the embodiment illustrated in FIG. **19** may determine a minimum largest chunk when the longest match does not contain a delimiter, and a maximum smallest chunk when the longest match does contain a delimiter.

(189) Referring to FIG. **19**, the method may start at operation **1979-1**. At operation **1979-2**, which may be the beginning of a repeat sequence, the method may initialize a chunk size (c) and/or an early termination bit (e), for example, to zero. At operation **1979-3**, the method may initialize a pointer P and a length L. At operation **1979-4**, the method may look for a longest match (c, L) in a lookahead buffer.

(190) At operation **1979-5**, if a longest match is found, the method may take a first major branch down the left side of FIG. **19** as follows. At operation **1979-6**, the method may set the pointer P to the beginning of the longest match and the length L to the length of the longest match. At operation **1979-7** the method may determine if the length of the current chunk size is greater than or equal to a default chunk size N (e.g., $c+L \geq N$). If the current chunk size is not greater than or equal to the default chunk size N, the method may proceed to operation **1979-11** as described below. If, however, the current chunk size is greater than or equal to the default chunk size, the method may initialize a record delimiter length (l) at operation **1979-8**. At operation **1979-9**, if the method does not find a first delimiter match (c, l) in the lookahead buffer, the method may proceed to operation **1979-11** as described below. If, however, at operation **1979-9**, the method finds a first delimiter match (c, l) in the lookahead buffer, the method may proceed to operation **1979-10** where the method may set the early termination bit to one ($e=1$), set the pointer P to the beginning of the first delimiter match in a sliding window, and/or set the length L to the length of the first delimiter match. At operation **1979-11**, the method may output the pointer P and the length L. The method may proceed to a common branch down the right side of FIG. **19** beginning with operation **1979-18**.

(191) At operation **1979-5**, if a longest match is not found, the method may take a second major branch down the center of FIG. **19** as follows. At operation **1979-12**, the method may set the pointer P to the value of a null pointer and the length L to 1. At operation **1979-13**, the method may output the null pointer P and the first byte of the lookahead buffer. At operation **1979-14**, the method may determine if the length of the current chunk size is greater than or equal to a default chunk size N (e.g., $c+L \geq N$). If the current chunk size is not greater than or equal to the default chunk size N, the method may proceed to the common branch down the right side of FIG. **19** beginning with operation **1979-18**. If, however, at operation **1979-14**, the current chunk size is

greater than or equal to the default chunk size, the method may initialize a record delimiter length (l) at operation **1979-15**. At operation **1979-16**, if the method does not find a first delimiter match (c, l) in the lookahead buffer, the method may proceed to the common branch down the right side of FIG. **19** beginning with operation **1979-18**. If, however, at operation **1979-16**, the method finds a first delimiter match (c, l) in the lookahead buffer, the method may proceed to operation **1979-17** where the method may set the early termination bit to one (e=1). The method may proceed to the common branch down the right side of FIG. **19** beginning with operation **1979-18**.

(192) The common branch down the right side of FIG. **19** may proceed as follows. At operation **1979-18**, the chunk size c may be updated by adding the length L (e.g., $c=c+L$), where L may be, for example, the length of the longest match, the length of the first delimiter match, or 1. At operation **1979-19**, the coding position and window may be moved a distance L (e.g., L bytes). At operation **1979-20**, if the early termination bit is not set (e=0), the method may return to operation **1979-3** to begin another iteration of the repeat sequence. If, however, the early termination bit is determined to be set (e=1), the method may indicate that the end of the current chunk has been found at operation **1979-21**. At operation **1979-22**, the method may determine if the lookahead buffer is empty. If the lookahead buffer is not empty, the method may proceed to operation **1979-23** where the early termination bit e may be reset to zero, and the method may return to operation **1979-3** to begin another iteration of the repeat sequence. However, if at operation **1979-22** the lookahead buffer is empty, the method may terminate at operation **1979-24**.

(193) Although the embodiments illustrated in FIG. **13**, FIG. **14**, FIG. **15**, FIG. **16**, FIG. **17**, FIG. **18**, and/or FIG. **19** may be described in the context of input data streams and/or stream-based compression algorithms, the principles may also be applied to any other type of input and/or compression algorithms such as batch-based (e.g., file-based) input data and/or compression algorithms based on blocks, files, and/or the like.

(194) Table 2 illustrates an example embodiment of pseudo code for an integrated chunking and compression method in accordance with example embodiments of the disclosure. The embodiment illustrated in Table 2 may be used, for example, with either or both of the embodiments illustrated in FIG. **15** and/or FIG. **17**. In some embodiments, the method illustrated in Table 2 may implement a minimum largest chunk and/or a maximum smallest chunk. For example, depending on the implementation details, the embodiment illustrated in Table 2 may determine a minimum largest chunk when the longest match does not contain a delimiter, and a maximum smallest chunk when the longest match does contain a delimiter.

(195) TABLE-US-00002

TABLE 2	1 Start {	2 Initialize c, e	3 Repeat {	4 Initialize P,
L	5 Find longest match (c, L) in lookahead buffer	6 if longest match is found {	7	
	P = beginning of longest match in window	8	L = length of longest match	9
	If $c+L \geq N$ {	10 Initialize l	11 Find the 1.sup.st delimiter match (c, l) in	
	lookahead buffer	12	If 1.sup.st delimiter match is found {	13 Set e=1
	P = beginning of 1.sup.st delimiter match in window	14	L = length of	
	1.sup.st delimiter match (l)	15	} 16	17
	Else {	21 Set L=1, P=NULL	22	23
	lookahead buffer	23	If $c+L \geq N$ {	24
	delimiter match (c, l) in lookahead buffer	25	Set e=1	26
	Set e=1	27	} 28	29
	position and window L bytes	30	If e = 1 {	31
	empty	32	End of chunk	33
	End	34	If lookahead buffer	35
	Else	36	Initialize e	37
		38	} 39	40
			} // end of repeat	41
	end of start			

(196) FIG. **20** illustrates an embodiment of a system with integrated chunking and compression in accordance with example embodiments of the disclosure. The system illustrated in FIG. **20** may be used, for example, to implement any of the techniques described with respect to FIG. **13**, FIG. **14**, FIG. **15**, FIG. **16**, FIG. **17**, FIG. **18**, and/or FIG. **19**

(197) Referring to FIG. 20, the system may include a host 2024 and a computational storage device 2008 that may communicate through a connection 2022. The host 2024 may include content-aware data chunking and compression logic 2090 (which may be referred to as chunking and compression logic, integrated chunking and compression logic, or integrated logic). The chunking and compression logic may be configured to provide one or more chunks of data to the storage device 2008. For example, the data chunking and compression logic 2090 may divide an object or other original data into one or more chunks of data prior to compression so the storage device 2008 may decompress a chunk of data and perform an operation on the decompressed chunk of data.

(198) Because the chunking and compression logic 2090 may be content aware, in some embodiments, it may divide data into chunks based on boundaries between records. For example, the chunking and compression logic 2090 may look for a record delimiter in the data to divide the data into variable sized chunks such that a chunk may end with a record delimiter, and therefore, end with a complete record.

(199) In some embodiments, the chunking and compression logic 2090 may include one or more lookahead buffers, window buffers, and/or the like, and may implement any of the techniques described with respect to FIG. 13, FIG. 14, FIG. 15, FIG. 16, FIG. 17, FIG. 18, and/or FIG. 19

(200) The computational storage device 2008 may include decompression logic 2028, one or more processing elements 2029, and storage media 2030. The decompression logic 2028 may be configured to decompress a chunk of compressed data to a form on which the one or more processing elements 2029 may perform an operation. The one or more processing elements 2029 may be configured to perform any type of operation such as data selection (e.g., scanning, filtering, and/or the like), compute acceleration, graph processing, graphics processing, machine learning, and/or the like. The storage media 2030 may be used to store any data including one or more modified chunks of data sent by the host 2024.

(201) In some embodiments, the decompression logic 2028 and/or one or more processing elements 2029 may be configured to read and decompress one or more chunks of data from the storage media 2030 and return a specified subset of the data, or perform any other operation on the restored chunk of data, in response to a request which may include a query (e.g., an expression) received at the storage device 2008.

(202) In some embodiments, the computational storage device 2008 may further include indication logic that may be configured to provide one or more indications to the content-aware data chunking and compression logic 2090. The one or more indications may include information that may be used to determine how to divide original data into chunks as described above with respect to the embodiment illustrated in FIG. 5.

(203) The host 2024 may be implemented with any component or combination of components as described above with respect to the embodiment illustrated in FIG. 4. The storage device 2008 may be implemented in any form factor, using any connector configuration, and any storage media as described above with respect to the embodiment illustrated in FIG. 4. The host 2024 and storage device 2008 may communicate using any type of storage interface and/or protocol, and may be implemented entirely or partially with, and/or used in connection with, a server chassis, server rack, dataroom, datacenter, edge datacenter, mobile edge datacenter, and/or any combinations thereof as described above with respect to the embodiment illustrated in FIG. 4. The communication connection 2022 may be implemented with any interconnect and/or network interfaces and/or protocols as described above with respect to the embodiment illustrated in FIG. 4. Any of the functionality, including any of the logic such as the content-aware data chunking and compression logic 2090, the decompression logic 2028, the one or more processing elements 2029, and/or the like, may be implemented with hardware, software or a combination thereof as described above with respect to the embodiment illustrated in FIG. 4. In some embodiments, one or more of the decompression logic 2028, processing elements 2029, and/or the like may be integrated with one or more other components of a storage device such as a storage device controller, a flash translation

layer (FTL) and/or the like. The content-aware data chunking and compression logic **2090** and the decompression logic **2028** may implement any compression and/or decompression techniques such as LZ77, gzip, Snappy, and/or the like.

(204) FIG. **21** illustrates an embodiment of a content-aware scheme for dividing data into chunks and encrypting data in accordance with example embodiments of the disclosure. In the embodiment illustrated in FIG. **21**, unencrypted input data **2168** (which may be stored in a buffer) shown at the top of FIG. **21** may include one or more records that may each terminate with a delimiter which, in this example, may be a semicolon. In other embodiments, however, other delimiters such as brackets, carriage return-line feed (CR-LF) sequences, and/or the like, may be used. In some embodiments, one or more records may be included in an array (e.g., a JSON array), a document (e.g., a JSON document), and/or the like within the input data **2168**. If a record ends at the end of an array, a document, and/or the like, a different delimiter may be used to indicate the end of the array, document, and/or the like, but the embodiment illustrated in FIG. **21** may also interpret the different delimiter as the end of a record.

(205) During a first pass (Operation 1), the method may scan the input data **2168** and use it to generate a chunk of data until it reaches a default chunk size N. If the last data element in the chunk **2169** is a delimiter, the method may proceed to a second pass (Operation 2). If the last data element in the chunk is not a delimiter, the method may modify the size of the chunk so the end of the chunk aligns with the end of a delimiter. For example, the method may begin at position N+1 in a buffer holding the data **2168** and scan backward until it finds the delimiter **2170**. The method may reduce the size of the chunk (e.g., shrink the chunk) until the delimiter **2170** is the last data element in the chunk. Thus, the chunk may be reduced by a size equal to the shrink space S so the chunk may only include one or more complete records.

(206) Alternatively, the method may begin at position N+1 in a buffer holding the data **2168** and scan forward until it finds the delimiter **2171**. The method may extend the size of the chunk until the delimiter **2171** is the last data element in the chunk **2169** as shown in the middle of FIG. **21**. Thus, the chunk may be extended by a size equal to the extended space E so the chunk may only include one or more complete records.

(207) In some embodiments, once the chunk size is determined, the method may proceed to a second pass (Operation 2) in which the self-contained chunk (e.g., chunk **2169**), which may end with a delimiter and only include complete records, may be encrypted using one or more encryption algorithms to generate encrypted data **2172**.

(208) In some embodiments, because the size of a chunk of unencrypted data may be reduced or extended, the size of the chunk of unencrypted data **2169** may be variable, and thus, the size of the chunk of encrypted data **2172** may be variable.

(209) Whether the method reduces or extends the length of the chunk may depend on various factors such as a maximum and/or optimal data size for an object or key-value storage device the chunk may be sent to, or a block size for a block-based storage device the chunk may be sent to. In the case of an object or key-value storage device, the resulting chunk of encrypted, self-contained data **2172** may be stored without further modification and/or processing.

(210) Depending on the implementation details, the content-aware data chunking embodiment illustrated in FIG. **21** may improve the operation of a storage system such as that shown in FIG. **11**, for example, because fewer or no records may be split between chunks on different storage devices, and thus, fewer or no fragments of records may be transferred between storage devices, storage nodes, storage servers, and/or the like, to enable components to restore chunks of data having complete records and/or perform operations on complete records.

(211) The encryption scheme illustrated in FIG. **21** may be implemented, for example, using any of the systems illustrated in FIG. **4**, FIG. **5**, and/or FIG. **9**, in which the chunking logic **426**, **526**, and/or **826** may include content-aware logic configured to determine a chunk size based on finding boundaries (e.g., delimiters) of records or other units of data in the unencrypted input data **2168**.

(212) Although the embodiment illustrated in FIG. 21 may improve the operation of a storage system, in some embodiments, and depending on the implementation details, the overhead associated with the use of two passes may increase the processing burden, especially when scanning relatively large objects.

(213) Some additional content-aware data chunking techniques in accordance with example embodiments of the disclosure may integrate a chunking operation with an encryption operation that may scan the data to be chunked. Depending on the implementation details, this may improve the efficiency of the chunking operation because it may exploit a scanning operation that was already being performed for purposes of encryption. Thus, in some embodiments, it may reduce or eliminate overhead associated with two scanning passes.

(214) FIG. 22 illustrates an example embodiment of a block cipher scheme that may be used to encrypt data techniques in accordance with example embodiments of the disclosure. The embodiment illustrated in FIG. 22 may be based, for example, on an AES algorithm. Data to be encrypted may be received as blocks (e.g., 128-bit blocks) of plaintext input data **2291**, processed in a series of linked operations including a pre-round transformation **2292**, one or more AES encryption rounds **2293-1**, . . . , **2293-N.sub.r** using round keys $K_{\text{sub}.0}$, $K_{\text{sub}.1}$, . . . , $K_{\text{sub}.R}$ generated by key expansion logic **2296**, and output as corresponding blocks (e.g., 128-bit blocks) of ciphertext output data **2294**. The number of rounds R may be determined, for example, by the length of a cipher key **2295** (e.g., 128, 192, or 256 bits) used for the encryption rounds. In some embodiments, the size of the blocks of ciphertext output data **2294** may be the same size as the blocks of plaintext input data **2291**.

(215) In some embodiments, the terms plaintext and/or ciphertext may refer to not just to actual text, but to any data in any form such as images, audio, video, and/or the like.

(216) Because input data may be scanned for encryption with a block cipher scheme such as that illustrated in FIG. 22, the embodiment illustrated in FIG. 22 may be integrated with a content-aware data chunking technique in accordance with example embodiments of the disclosure. Depending on the implementation details, this may improve the efficiency of the chunking operation because it may exploit a scanning operation that may already be performed for purposes of encryption. Thus, in some embodiments, it may reduce or eliminate overhead associated with two scanning passes because scanning for encryption and scanning for a delimiter may be performed at the same time.

(217) FIG. 23 illustrates an embodiment of an integrated data chunking and encryption scheme in accordance with example embodiments of the disclosure. FIG. 24 illustrates an embodiment of an integrated data chunking and encryption method in accordance with example embodiments of the disclosure. The embodiment illustrated in FIG. 24 may be used, for example, with the scheme illustrated in FIG. 23.

(218) In the scheme illustrated in FIG. 23, a stream of unencrypted input data **2368** shown at the top of FIG. 23 may include one or more records that may each terminate with a delimiter which, in this example, may be a semicolon. In other embodiments, however, other delimiters such as brackets, carriage return-line feed (CR-LF) sequences, and/or the like, may be used. In some embodiments, one or more records may be included in an array (e.g., a JSON array), a document (e.g., a JSON document), and/or the like within the input data **2368**. If a record ends at the end of an array, a document, and/or the like, a different delimiter may be used to indicate the end of the array, document, and/or the like, but the method illustrated in FIG. 23 may also interpret the different delimiter as the end of a record.

(219) The scheme illustrated in FIG. 23 may be implemented with one or more of the following operations as illustrated in FIG. 24.

(220) (1) At a first operation **2497-1**, the method may scan the stream of input data **2368** and apply a block cipher, such as that illustrated in FIG. 22, to one or more portions (which may also be referred to as parts) **2398** of the stream of input data **2368** as shown in FIG. 23. The block cipher

may be applied to the stream of input data **2368** on a block-by-block basis in which the size of the one or more parts **2398** may correspond to the block size used by the block cipher, which in turn, may correspond to a key length used by the block cipher. The stream of input data **2368** may be encrypted using the block cipher, for example, until a default chunk size N is reached. In some embodiments, the default chunk size N may be set to an even multiple of the block size (e.g. a key length) used by the block cipher.

(221) (2) At a second operation **2497-2**, if the last data element of the current chunk of data is not a delimiter, the method may continue scanning the stream of input data **2368** to find a delimiter **2399** and expand the chunk of data by an amount E until the delimiter **2399** is the last data element of the chunk as shown at the top of FIG. **23**.

(222) (3) At a third operation **2497-3**, if the end of the chunk of data does not align with the end of a block used by the block cipher, the method may pad the chunk of data with padding elements “p” to further extend the chunk **23100** by an amount p such that the end of the chunk aligns with the end of a block as shown at the bottom of FIG. **23**.

(223) (4) At a fourth operation **2497-4**, if the chunk has been extended, the block cipher may be applied to the block including the parts E and p (if any).

(224) (5) At a fifth operation **2497-5**, the resulting encrypted chunk of data **23101**, which may have a variable length based on the position of the delimiter **2399** in the stream of input data, may be stored, for example, in a KV drive.

(225) In some embodiments, the embodiments illustrated in FIG. **23** and/or FIG. **24** may be continued, for example, until the end of a user object in the stream of input data is reached. An object may be divided and encrypted into one or more chunks of encrypted data. Because (a) the size of the plaintext input data may be variable, (b) different default chunk sizes N may be used, and/or (c) a chunk may be extended to include a delimiter and/or one or more padding elements, the size of a chunk of encrypted data may be variable. In the case of an object or key-value storage device, the resulting chunk of encrypted, self-contained data may be stored without further modification and/or processing.

(226) Depending on the implementation details, the content-aware data chunking embodiments illustrated in FIG. **23** and/or FIG. **24** may improve the operation of a storage system such as that shown in FIG. **11**, for example, because fewer or no records may be split between chunks on different storage devices, and thus, fewer or no fragments of records may be transferred between storage devices, storage nodes, storage servers, and/or the like, to enable components to restore chunks of data having complete records and/or perform operations on complete records. Moreover, because the embodiments illustrated in FIG. **23** and/or FIG. **24** may exploit a scanning operation that was already being performed for purposes of encryption to perform data chunking, it may reduce or eliminate overhead associated with two scanning passes.

(227) For purposes of illustration, the embodiments illustrated in FIG. **22**, FIG. **23**, and/or FIG. **24** may be described in the context of block ciphers such as AES. However, the principles may also be applied to schemes using stream ciphers such as RC4.

(228) FIG. **25** illustrates an embodiment of a system with integrated chunking and encryption in accordance with example embodiments of the disclosure. The system illustrated in FIG. **25** may be used, for example, to implement any of the techniques described with respect to FIG. **22**, FIG. **23**, and/or FIG. **24**.

(229) Referring to FIG. **25**, the system may include a host **2524** and a computational storage device **2508** that may communicate through a connection **2522**. The host **2524** may include content-aware data chunking and encryption logic **25101** (which may be referred to as chunking and encryption logic, integrated chunking and encryption logic, or integrated logic). The chunking and encryption logic may be configured to provide one or more chunks of data to the storage device **2508**. For example, the data chunking and encryption logic **25101** may divide an object or other original data into one or more chunks of data prior to encryption so the storage device **2508** may decrypt a chunk

of data and perform an operation on the decrypted chunk of data.

(230) Because the chunking and encryption logic **25101** may be content aware, in some embodiments, it may divide data into chunks based on boundaries between records. For example, the chunking and encryption logic **25101** may look for a record delimiter in the data to divide the data into variable sized chunks such that a chunk may end with a record delimiter, and therefore, end with a complete record.

(231) The computational storage device **2508** may include decryption logic **2528**, one or more processing elements **2529**, and storage media **2530**. The decryption logic **2528** may be configured to decrypt a chunk of encrypted data to a form on which the one or more processing elements **2529** may perform an operation. The one or more processing elements **2529** may be configured to perform any type of operation such as data selection (e.g., scanning, filtering, and/or the like), compute acceleration, graph processing, graphics processing, machine learning, and/or the like. The storage media **2530** may be used to store any data including or more modified chunks of data sent by the host **2524**.

(232) In some embodiments, the decryption logic **2528** and/or one or more processing elements **2529** may be configured to read and decrypt one or more chunks of data from the storage media **2530** and return a specified subset of the data, or perform any other operation on the restored chunk of data, in response a request which may include a query (e.g., an expression) received at the storage device **2508**.

(233) In some embodiments, the computational storage device **2508** may further include indication logic that may be configured to provide one or more indications to the content-aware data chunking and decryption logic **25101**. The one or more indications may include information that may be used to determine how to divide original data into chunks as described above with respect to the embodiment illustrated in FIG. 5.

(234) The host **2524** may be implemented with any component or combination of components as described above with respect to the embodiment illustrated in FIG. 4. The storage device **2508** may be implemented in any form factor, using any connector configuration, and any storage media as described above with respect to the embodiment illustrated in FIG. 4. The host **2524** and storage device **2508** may communicate using any type of storage interface and/or protocol, and may be implemented entirely or partially with, and/or used in connection with, a server chassis, server rack, dataroom, datacenter, edge datacenter, mobile edge datacenter, and/or any combinations thereof as described above with respect to the embodiment illustrated in FIG. 4. The communication connection **2522** may be implemented with any interconnect and/or network interfaces and/or protocols as described above with respect to the embodiment illustrated in FIG. 4. Any of the functionality, including any of the logic such as the content-aware data chunking and encryption logic **25101**, the decryption logic **2528**, the one or more processing elements **2529**, and/or the like, may be implemented with hardware, software or a combination thereof as described above with respect to the embodiment illustrated in FIG. 4. In some embodiments, one or more of the decryption logic **2528**, processing elements **2529**, and/or the like may be integrated with one or more other components of a storage device such as a storage device controller, a flash translation layer (FTL) and/or the like. The content-aware data chunking and encryption logic **25101** and the decryption logic **2528** may implement any encryption and/or decryption techniques such as Data Encryption Standard (DES), AES, and/or the like.

(235) Although some embodiments may be described in the context of integrated data chunking and compression, encryption, and/or the like for use by computational storage devices, the principles are not limited to any particular application and may be applied for use in any data chunking applications, and for use with any types of devices including accelerators, network interface cards (NICs), and/or the like.

(236) For purposes of illustration, embodiments that may implement integrated chunking and compression and integrated chunking and encryption may be shown separately, but the principles

may be combined in accordance with example embodiments of the disclosure to implement schemes that may implement integrated data chunking, compression, and encryption.

(237) The principles disclosed herein may be applied to integrate data chunking, not only into data compression and/or data encryption operations, but also into any type of operation that may scan data including, for example, filtering, erasure coding, searching, machine learning, graph processing, and/or the like. For example, in the embodiments illustrated in FIG. 20 and FIG. 25, the data chunking and compression logic 2090 and/or the data chunking and encryption logic 25101 may be adapted to integrate scanning, for purposes of chunking, with scanning for any other operation. Thus, in some embodiments, a method may include scanning input data, performing, based on the scanning, an operation using the input data to generate processed data, finding, based on the scanning, a delimiter in the input data, and generating, based on a position of the delimiter in the input data, a chunk of data using the processed data. In some embodiments, the data may include a stream of input data, and the operation may be a streaming-based operation. In some embodiments, the operation may be a block-based operation. In some embodiments, a system may include a host including host logic configured to perform a scanning operation on input data, perform, based on the scanning operation, a processing operation to generate processed data using the input data, find, based on the scanning operation, a delimiter in the input data, and generate, based on the delimiter, a chunk of data using the processed data. In some embodiments, the system may further include a device configured to receive the chunk of data from the host, the device including device logic configured to restore the chunk of data to generate a restored chunk of data, and perform an operation on the restored chunk of data.

(238) FIG. 26 illustrates an example embodiment of a host apparatus for a storage scheme with data chunk modification in accordance with example embodiments of the disclosure. The host 2600 illustrated in FIG. 26 may be used to implement any of the host functionality disclosed herein. The host 2600 may be implemented with any component or combination of components such as one or more clients, one or more object storage servers, one or more storage nodes, and/or the like, or a combination thereof.

(239) The host apparatus 2600 illustrated in FIG. 26 may include a processor 2602, which may include a memory controller 2604, a system memory 2606, host control logic 2608, and/or communication interface 2610. Any or all of the components illustrated in FIG. 26 may communicate through one or more system buses 2612. In some embodiments, one or more of the components illustrated in FIG. 26 may be implemented using other components. For example, in some embodiments, the host control logic 2608 may be implemented by the processor 2602 executing instructions stored in the system memory 2606 or other memory.

(240) The host control logic 2608 may include and/or implement any of the host functionality disclosed herein including data chunking logic 426, 526, and/or 826, data modification logic 427 and/or 527, compression logic 846, encryption logic 847, erasure coding logic 848, data selection logic 849, cluster management logic 850, node and device management logic 851, processing units 852, content-aware data chunking and compression logic 2090, content-aware data chunking and decryption logic 25101, and/or the like.

(241) FIG. 27 illustrates an example embodiment of a storage device with data chunk modification in accordance with example embodiments of the disclosure. The storage device 2700 illustrated in FIG. 27 may be used to implement any of the storage device functionality disclosed herein. The storage device 2700 may include a device controller 2702, a media translation layer 2704, a storage media 2706, computational resources 2708, processing control logic 2716, and a communication interface 2710. The components illustrated in FIG. 27 may communicate through one or more device buses 2712. In some embodiments that may use flash memory for some or all of the storage media 2706, the media translation layer 2704 may be implemented partially or entirely as a flash translation layer (FTL).

(242) In some embodiments, the processing control logic 2716 may be used to implement any of

the storage device functionality disclosed herein including data restoration logic **428** and/or **528**, processing elements **429**, **529**, and/or **1429**, indication logic **531**, data selection engine **853**, decryption logic **854**, decompression logic **855**, data selection logic **856**, key-value logic **857**, decompression logic **2028**, one or more processing elements **2029**, decryption logic **2528**, the one or more processing elements **2529**, and/or the like.

(243) As mentioned above, any of the functionality described herein, including any of the host (e.g., client, storage server, storage node, and/or the like) functionality, storage device functionality, and/or the like, described herein may be implemented with hardware, software, or any combination thereof, including combinational logic, sequential logic, one or more timers, counters, registers, state machines, volatile memories such as DRAM and/or SRAM, nonvolatile memory and/or any combination thereof, CPLDs, FPGAs, ASICs, CPUs including CISC processors such as x86 processors and/or RISC processors such as ARM processors, GPUs, NPUs, and/or the like, executing instructions stored in any type of memory. In some embodiments, one or more components may be implemented as a system-on-chip (SOC).

(244) FIG. **28** illustrates an embodiment of a method for data compression in accordance with example embodiments of the disclosure. The method may begin at operation **2802**. At operation **2804**, the method may scan input data. In some embodiments, the input data may be received as a stream of input data. At operation **2806**, the method may perform, based on the scanning, a compression operation to generate compressed data using the input data. At operation **2808**, the method may find, based on the scanning, a delimiter in the input data. For example, the input data may include one or more records, and a delimiter may indicate a boundary of a record. At operation **2810**, the method may generate, based on a position of the delimiter in the input data, a portion of data using the compressed data. The method may end at operation **2812**.

(245) FIG. **29** illustrates an embodiment of a method for data encryption in accordance with example embodiments of the disclosure. The method may begin at operation **2902**. At operation **2904**, the method may scan input data. In some embodiments, the input data may be received as a stream of input data. At operation **2906**, the method may perform, based on the scanning, an encryption operation to generate encrypted data using the input data. At operation **2908**, the method may find, based on the scanning, a delimiter in the input data. For example, the input data may include one or more records, and a delimiter may indicate a boundary of a record. At operation **2910**, the method may generate, based on a position of the delimiter in the input data, a portion of data using the encrypted data. The method may end at operation **2912**.

(246) FIG. **30** illustrates an embodiment of a method for data dividing data into one or more portions data in accordance with example embodiments of the disclosure. The method may begin at operation **3002**. At operation **3004**, the method may scan input data. In some embodiments, the input data may be received as a stream of input data. At operation **3006**, the method may perform, based on the scanning, an operation on the input data to generate processed data. In some embodiments, the operation may be any operation that may scan data such as compression, encryption, filtering, erasure coding, searching, machine learning, graph processing, and/or the like. At operation **3008**, the method may find, based on the scanning, a delimiter in the input data. For example, the input data may include one or more records, and a delimiter may indicate a boundary of a record. At operation **3010**, the method may generate, based on a position of the delimiter in the input data, a portion of data using the processed data. The method may end at operation **3012**.

(247) The embodiments illustrated in FIG. **28**, FIG. **29**, and FIG. **30**, as well as all of the other embodiments described herein, are example operations and/or components. In some embodiments, some operations and/or components may be omitted and/or other operations and/or components may be included. Moreover, in some embodiments, the temporal and/or spatial order of the operations and/or components may be varied. Although some components and/or operations may be illustrated as individual components, in some embodiments, some components and/or operations shown separately may be integrated into single components and/or operations, and/or some

components and/or operations shown as single components and/or operations may be implemented with multiple components and/or operations.

(248) Some embodiments disclosed above have been described in the context of various implementation details, but the principles of this disclosure are not limited to these or any other specific details. For example, some functionality has been described as being implemented by certain components, but in other embodiments, the functionality may be distributed between different systems and components in different locations and having various user interfaces. Certain embodiments have been described as having specific processes, operations, etc., but these terms also encompass embodiments in which a specific process, operation, etc. may be implemented with multiple processes, operations, etc., or in which multiple processes, operations, etc. may be integrated into a single process, step, etc. A reference to a component or element may refer to only a portion of the component or element. For example, a reference to a block may refer to the entire block or one or more subblocks. The use of terms such as “first” and “second” in this disclosure and the claims may only be for purposes of distinguishing the things they modify and may not indicate any spatial or temporal order unless apparent otherwise from context. In some embodiments, a reference to a thing may refer to at least a portion of the thing, for example, “based on” may refer to “based at least in part on,” and/or the like. A reference to a first element may not imply the existence of a second element. The principles disclosed herein have independent utility and may be embodied individually, and not every embodiment may utilize every principle. However, the principles may also be embodied in various combinations, some of which may amplify the benefits of the individual principles in a synergistic manner.

(249) The various details and embodiments described above may be combined to produce additional embodiments according to the inventive principles of this patent disclosure. Since the inventive principles of this patent disclosure may be modified in arrangement and detail without departing from the inventive concepts, such changes and modifications are considered to fall within the scope of the following claims.

Claims

1. A method for data compression, the method comprising: receiving input data in a first buffer; finding a delimiter in the input data; generating, based on a position of the delimiter in the input data, a first portion of data using a part of the input data, wherein a size of the first portion of data is determined based on the position of the delimiter in the input data, the size of the first portion of data is used to determine a size of a second portion of data in a second buffer; and compressing the second portion of data.
2. The method of claim 1, wherein: the input data comprises a record; the delimiter indicates a boundary of the record; and the first portion of data comprises the record.
3. The method of claim 1, wherein the position of the delimiter is in the part of the input data.
4. The method of claim 3, wherein the generating the first portion of data comprises generating the first portion of data based on a subset of the part of the input data.
5. The method of claim 1, wherein: the part of the input data is a first part of the input data; and the position of the delimiter is in a second part of the input data.
6. The method of claim 5, wherein the generating the portion of data comprises generating the first portion of data based on the first part of the input data and the second part of the input data.
7. The method of claim 1, wherein a size of the part of input data is based on a default portion size.
8. The method of claim 1, wherein the second buffer is a window.
9. A method for data compression, the method comprising: scanning input data; performing, based on the scanning, a compression operation to generate compressed data using the input data; finding, based on the scanning, a delimiter in the input data, wherein a size of a first portion of data is determined based on a position of the delimiter in the input data, the size of the first portion of data

is used to determine a size of a second portion of data in a second buffer; and generating, based on the position of the delimiter in the input data, the first portion of data using the compressed data.

10. The method of claim 9, wherein the input data comprises a record; the delimiter indicates a boundary of the record; and the portion of data comprises the record.

11. The method of claim 9, wherein the generating comprises generating the first portion of data based on a portion size.

12. The method of claim 11, wherein the portion size is a default portion size.

13. The method of claim 11, wherein the portion size is based on a default portion size and a length of a delimiter in the input data.

14. The method of claim 9, further comprising setting an indication based on a size of the compressed data.

15. The method of claim 14, wherein the generating comprises generating the first portion of data based on the indication and the delimiter.

16. The method of claim 14, wherein the scanning comprises scanning the input data based on the indication.

17. The method of claim 9, wherein the scanning comprises scanning the input data based on the delimiter.

18. The method of claim 9, wherein the performing comprises performing the compression operation based on the delimiter.

19. A system comprising: a host comprising a memory for storing instructions; and a processor configured to execute the instructions to: perform a scanning operation on input data; perform, based on the scanning operation, a data compression operation to generate compressed data using the input data; find, based on the scanning operation, a delimiter in the input data; and generate, based on the delimiter, a first portion of data using the compressed data, wherein a size of the first portion of data is determined based on a position of the delimiter in the input data, the size of the first portion of data is used to determine a size of a second portion of data in a second buffer.

20. The system of claim 19, further comprising a device configured to receive the first portion of data from the host, the device comprising device logic configured to: decompress the first portion of data to generate a decompressed portion of data; and perform an operation on the decompressed portion of data.
